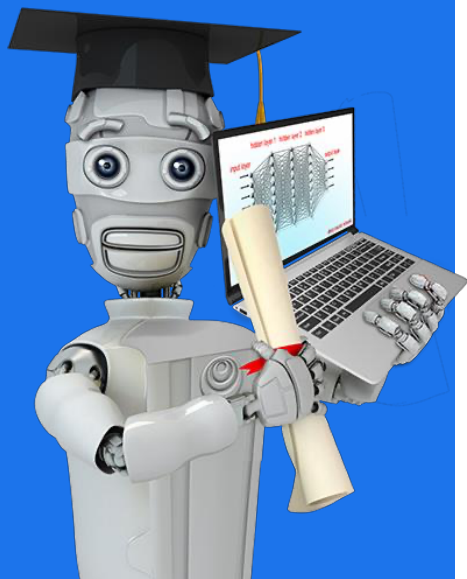


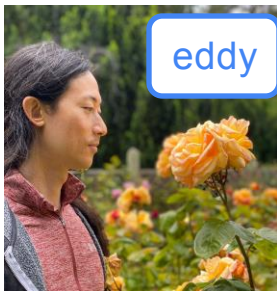
Stanford
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Machine Learning

Welcome!



eddy



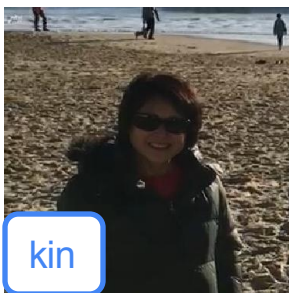
aarti



geoff



Ivy



kin



robert



andres



daniel

Re: Urgent Information :)

External

Spam x

Congratulations!
You've won
a million dollars!



Compose

Mail

Inbox

Starred

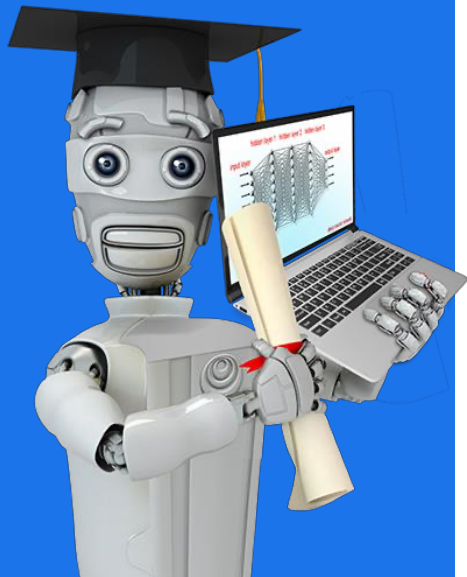
Snoozed

Sent

Drafts

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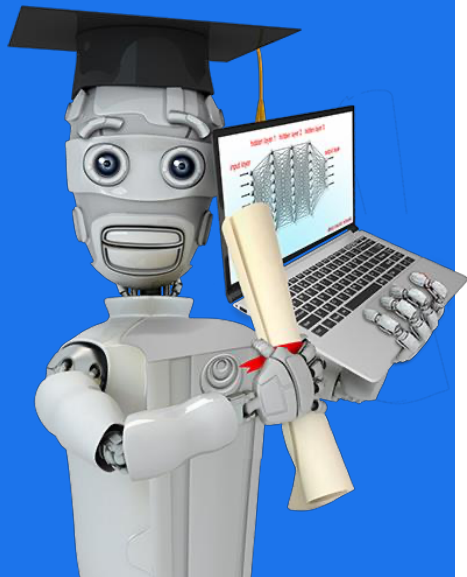


Machine Learning

Applications of Machine Learning

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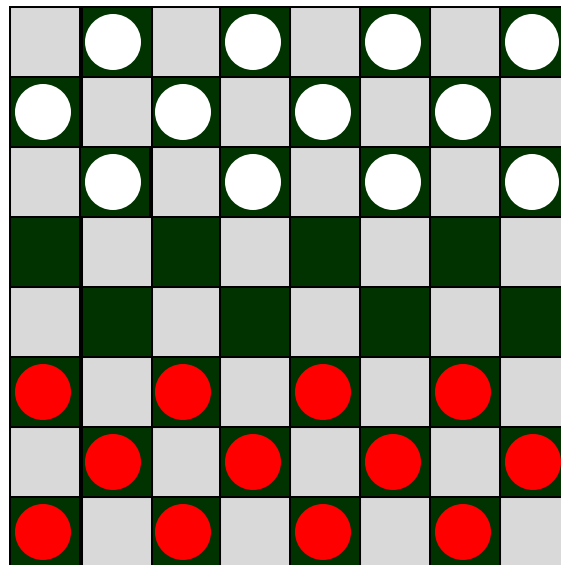
Machine Learning Overview

What is
Machine Learning?

Machine learning

“Field of study that gives computers the ability to learn without being explicitly programmed.”

Arthur Samuel (1959)



Question

If the checkers program had been allowed to play only ten games (instead of tens of thousands) against itself, a much smaller number of games, how would this have affected its performance?

☐ Would have made it better

→ ☒ Would have made it worse

Machine learning algorithms

rapid advancements

used most in real-world applications

- Supervised learning ← *course 1, 2*

- Unsupervised learning ←

course 3

- Recommender systems

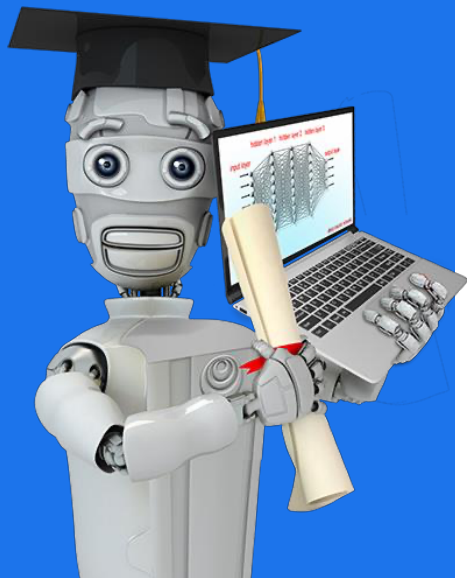
- Reinforcement learning

Practical advice for applying learning algorithms



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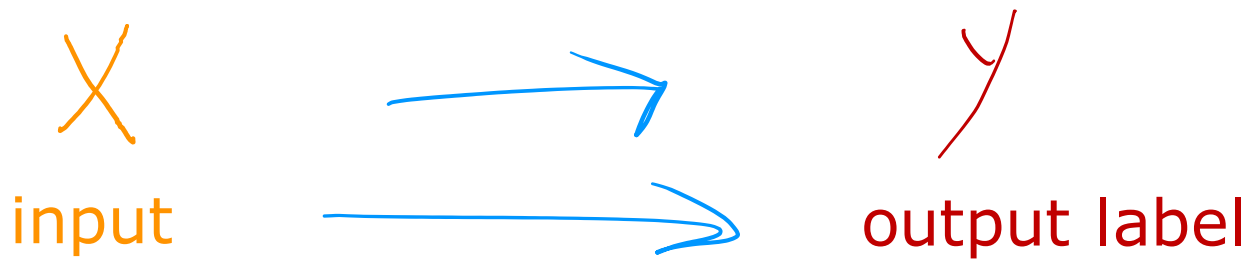
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Machine Learning Overview

Supervised Learning Part 1

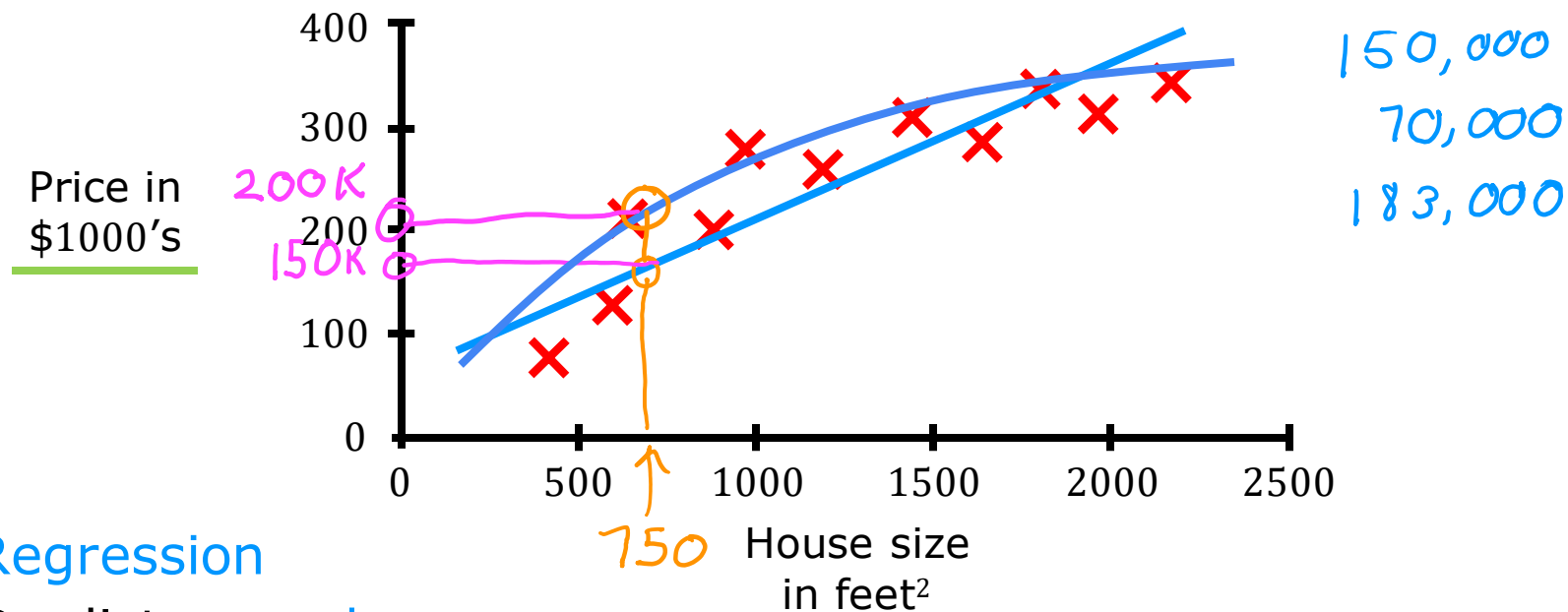
Supervised learning



Learns from being given “right answers”

Input (X)		Output (Y)	Application
email	→	spam? (0/1)	spam filtering
audio	→	text transcripts	speech recognition
English	→	Spanish	machine translation
ad, user info	→	click? (0/1)	online advertising
image, radar info	→	position of other cars	self-driving car
image of phone	→	defect? (0/1)	visual inspection

Regression: Housing price prediction



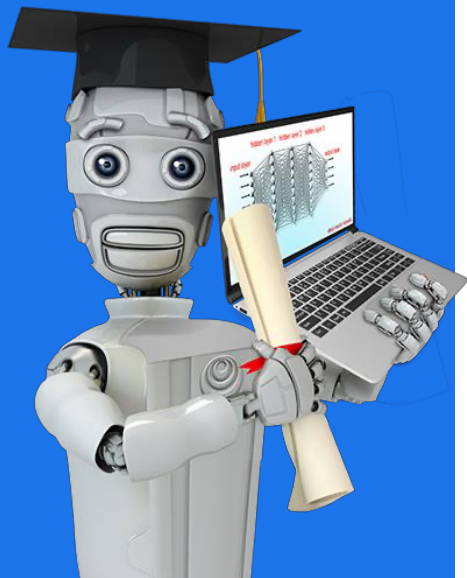
Regression

Predict a **number**

infinitely many possible outputs

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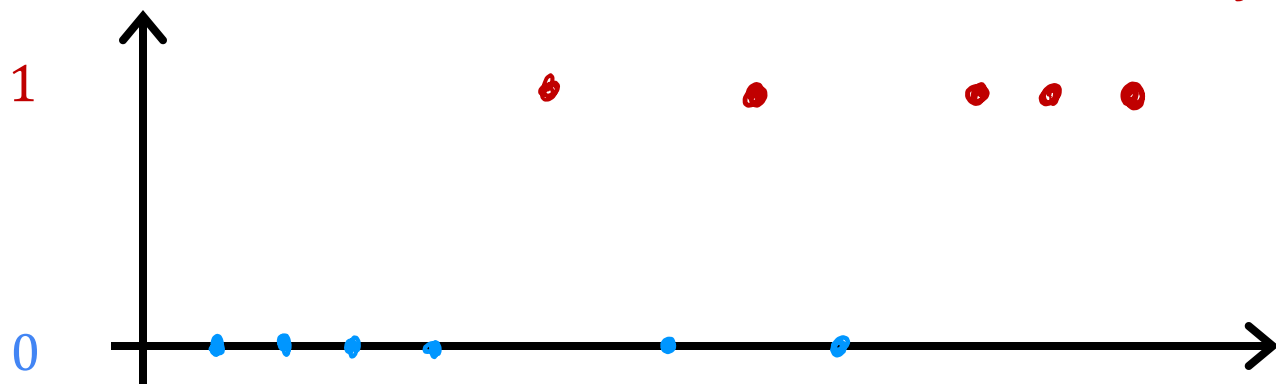
Machine Learning Overview

Supervised Learning Part 2

Classification: Breast cancer detection

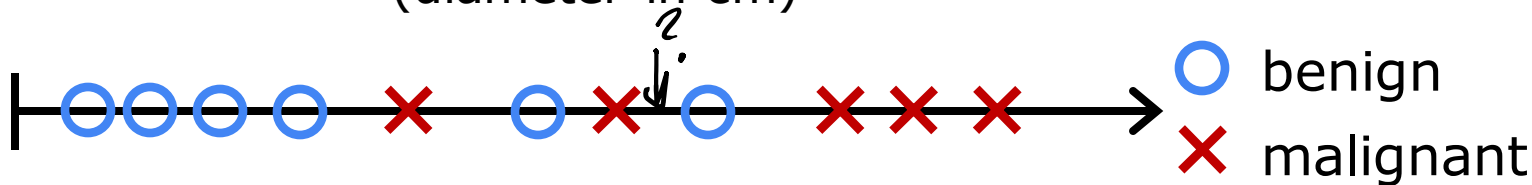


malignant benign



size	diagnosis
2	○
5	×
1	○
7	×
⋮	

tumor size x
(diameter in cm)



Classification: Breast cancer detection

- benign
- ✗ malignant *type 1*
- △ malignant type 2



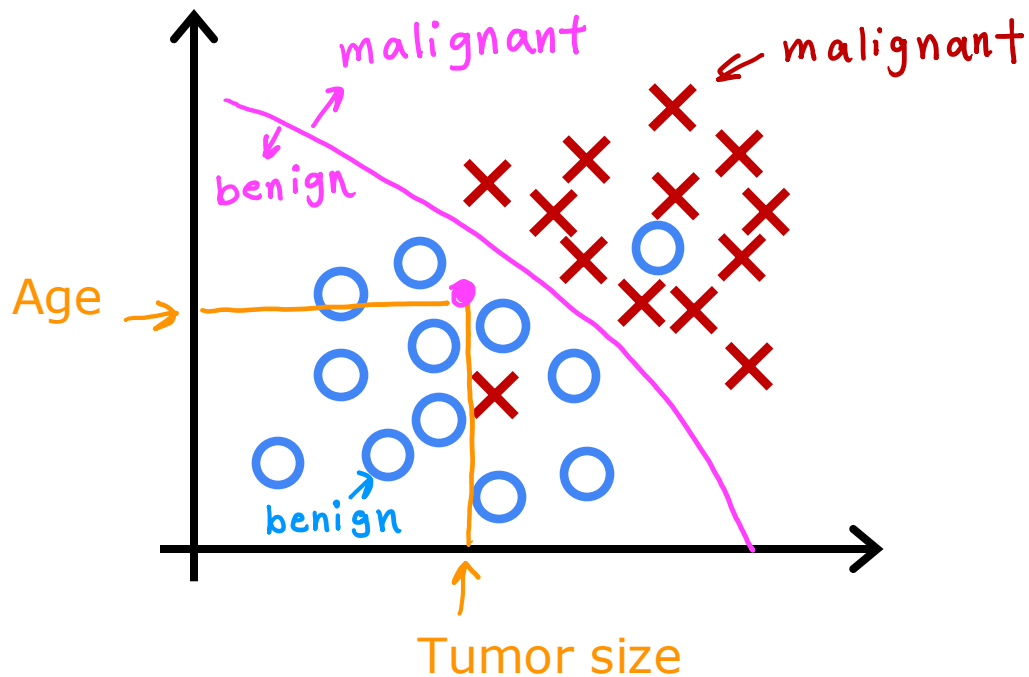
class category

Classification

predict *categories* *cat dog benign malignant 0, 1, 2*

small number of possible outputs

Two or more inputs



Supervised learning

Learns from being given “right answers”

Regression

Predict a number

infinitely many possible outputs

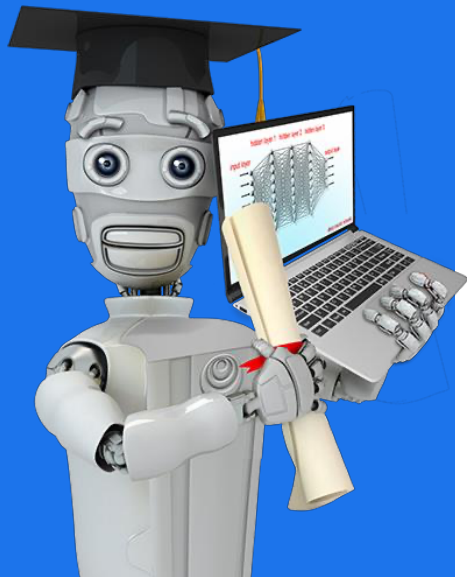
Classification

predict categories

small number of possible outputs

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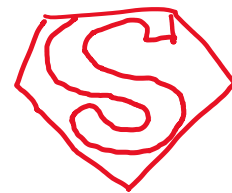


Machine Learning Overview

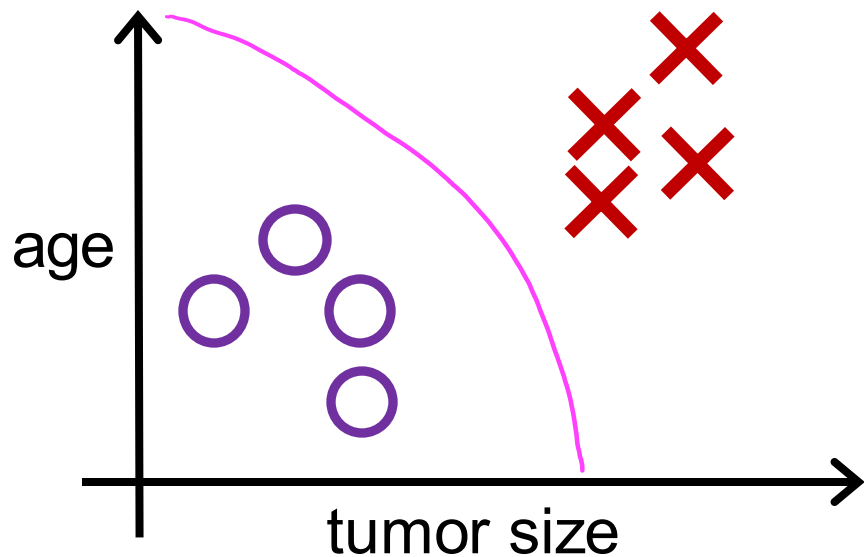
Unsupervised Learning Part 1

Previous: Supervised learning

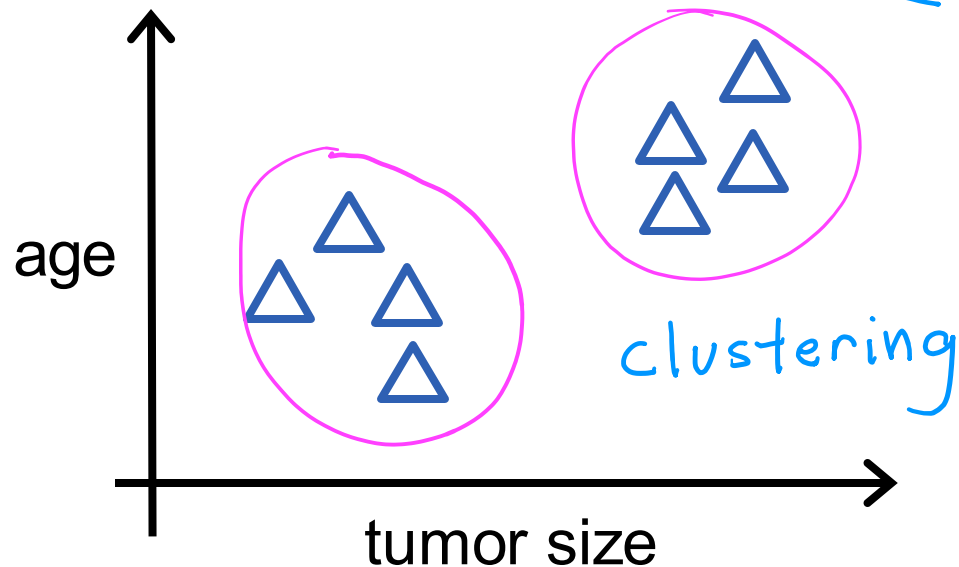
Now: Unsupervised learning



Supervised learning
Learn from data **labeled**
with the “**right answers**”



Unsupervised learning
Find something interesting
in **unlabeled** data.

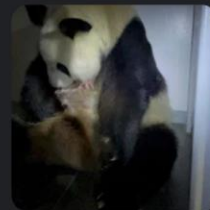


Clustering: Google news



Giant **panda** gives birth to rare **twin** cubs at Japan's oldest **zoo**

USA TODAY · 6 hours ago



• Giant **panda** gives birth to **twin** cubs at Japan's oldest **zoo**

CBS News · 7 hours ago

• Giant **panda** gives birth to **twin** cubs at Tokyo's Ueno **Zoo**

WHBL News · 16 hours ago

• A Joyful Surprise at Japan's Oldest **Zoo**: The Birth of **Twin Pandas**

The New York Times · 1 hour ago

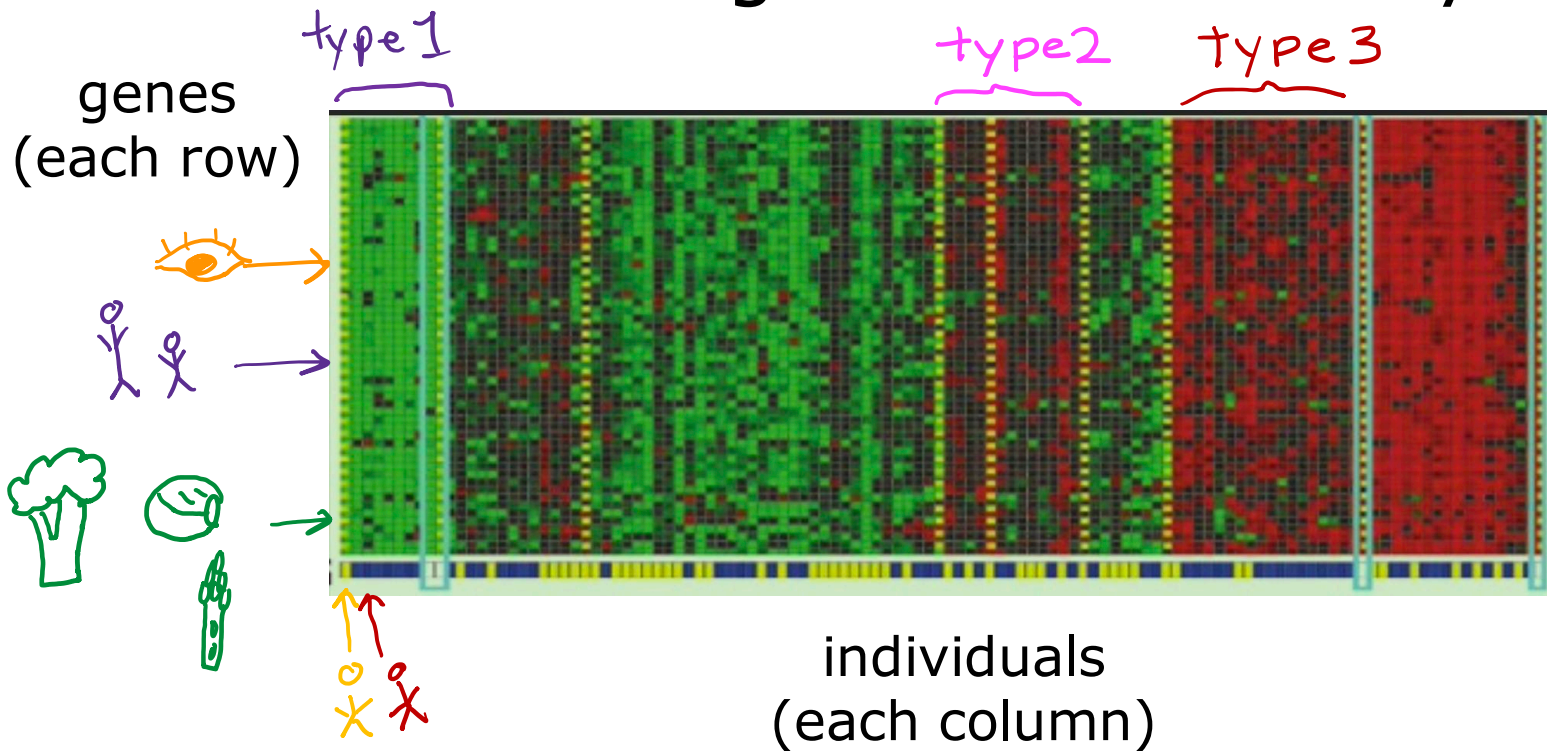
• **Twin Panda** Cubs Born at Tokyo's Ueno **Zoo**

PEOPLE · 6 hours ago

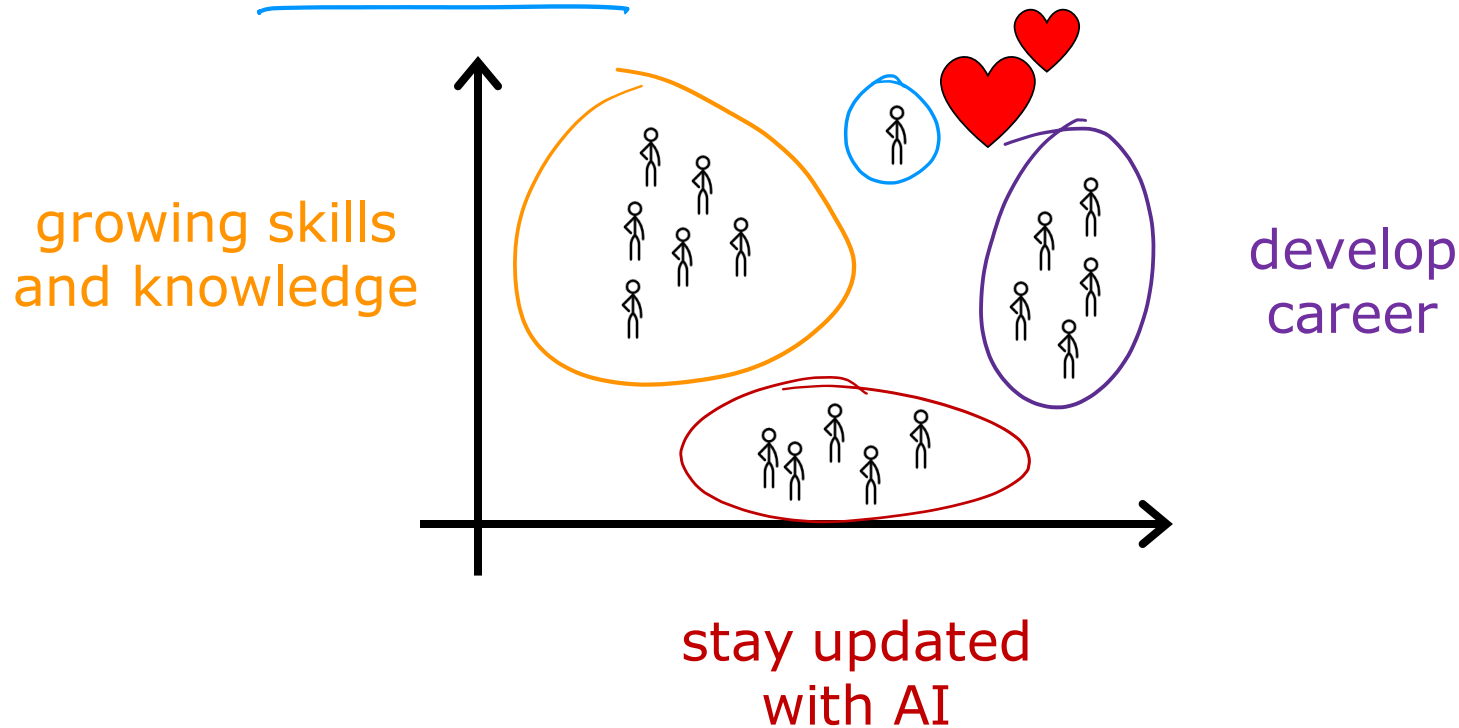
 [View Full Coverage](#)



Clustering: DNA microarray

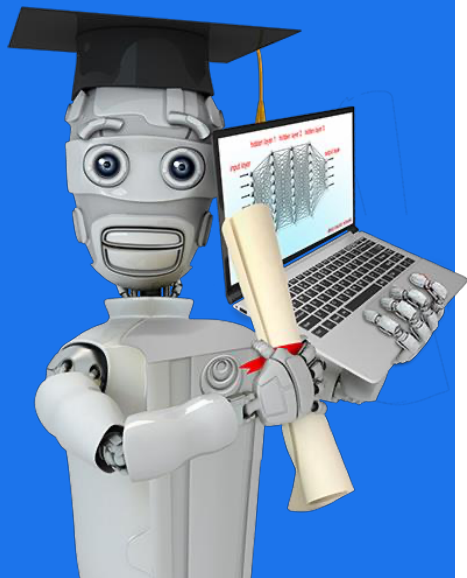


Clustering: Grouping customers



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Machine Learning Overview

Unsupervised Learning Part 2

Unsupervised learning

Data only comes with inputs x , but not output labels y .
Algorithm has to find **structure** in the data.

Clustering

Group similar data points together.

Dimensionality reduction

Compress data using fewer numbers.

Anomaly detection

Find unusual data points.

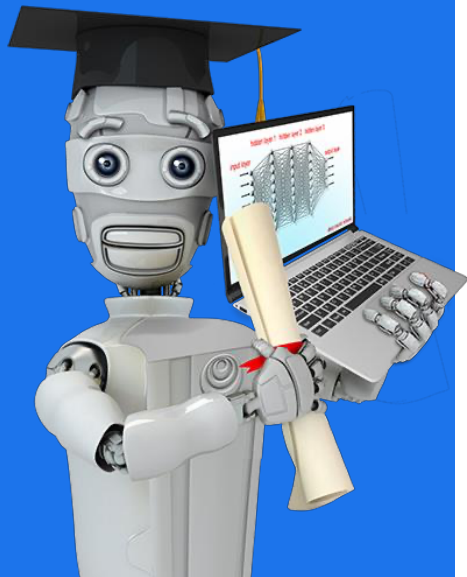
Question

Of the following examples, which would you address using an **unsupervised** learning algorithm?

-  ☐ Given email labeled as spam/not spam, learn a spam filter.
-  ☒ Given a set of news articles found on the web, group them into sets of articles about the same story.
-  ☒ Given a database of customer data, automatically discover market segments and group customers into different market segments.
-  ☐ Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not

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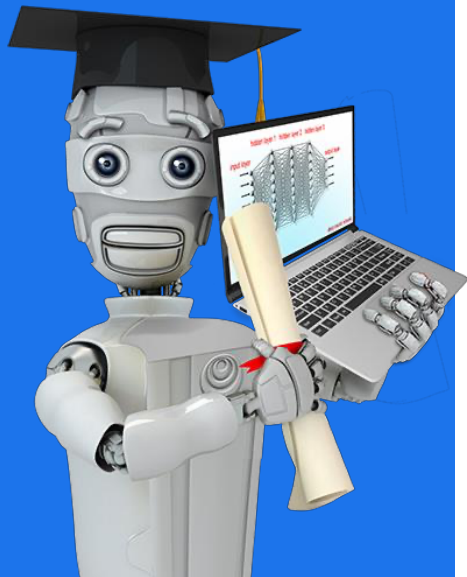


Machine Learning Overview

Jupyter Notebooks

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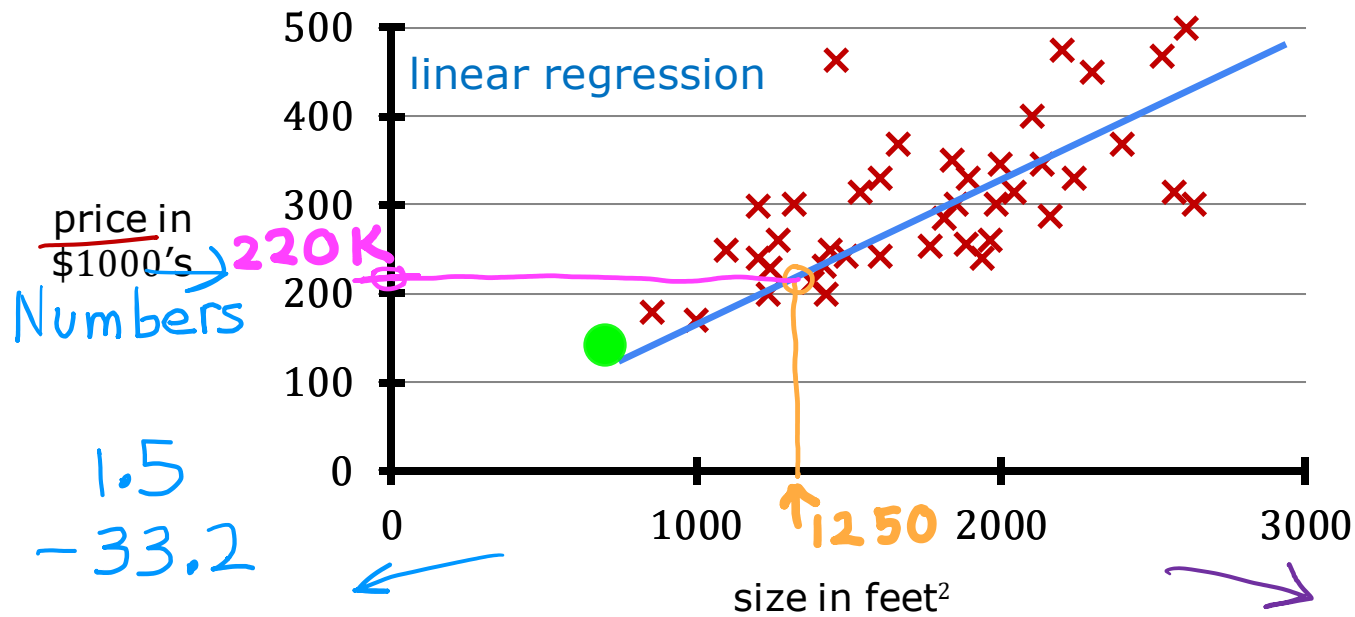
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Linear Regression with One Variable

Linear Regression Model Part 1

House sizes and prices



categories

cat } 2
dog }

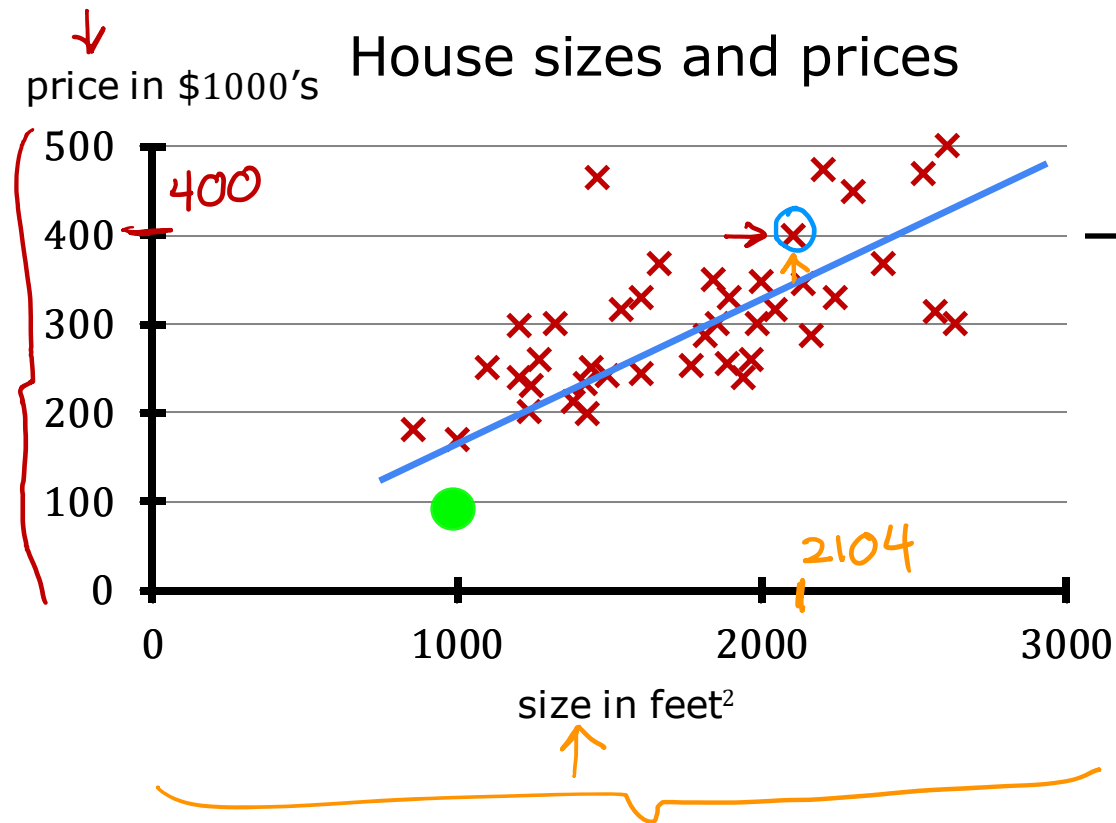
disease  10

Regression model
Predicts numbers
Infinitely many possible outputs

Supervised learning model
Data has "right answers"

Classification model
Predicts categories
Small number of possible outputs

House sizes and prices



Data table

size in feet ²	price in \$1000's
2104	400
1416	232
1534	315
852	178
...	...
3210	870

Terminology

Training Data used to train the model

set: x y
→ size in feet² → price in \$1000's

(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
...	...	...
(47)	3210	870

$m = 47$

$$x^{(1)} = 2104 \quad y^{(1)} = 400$$

$$(x^{(1)}, y^{(1)}) = (2104, 400)$$

$$x^{(2)} = 1416 \quad x^{(2)} \neq x^2 \text{ not exponent}$$

Notation:

x = "input" variable
feature

y = "output" variable
"target" variable

m = number of training examples

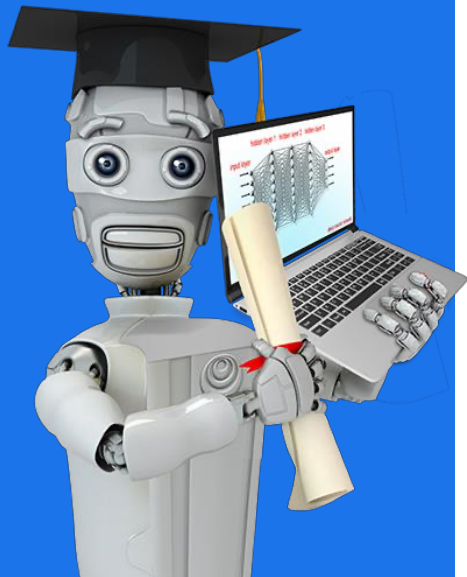
(x, y) = single training example

$$(x^{(i)}, y^{(i)})$$

$(x^{(i)}, y^{(i)})$ = i^{th} training example
index (1st, 2nd, 3rd ...)

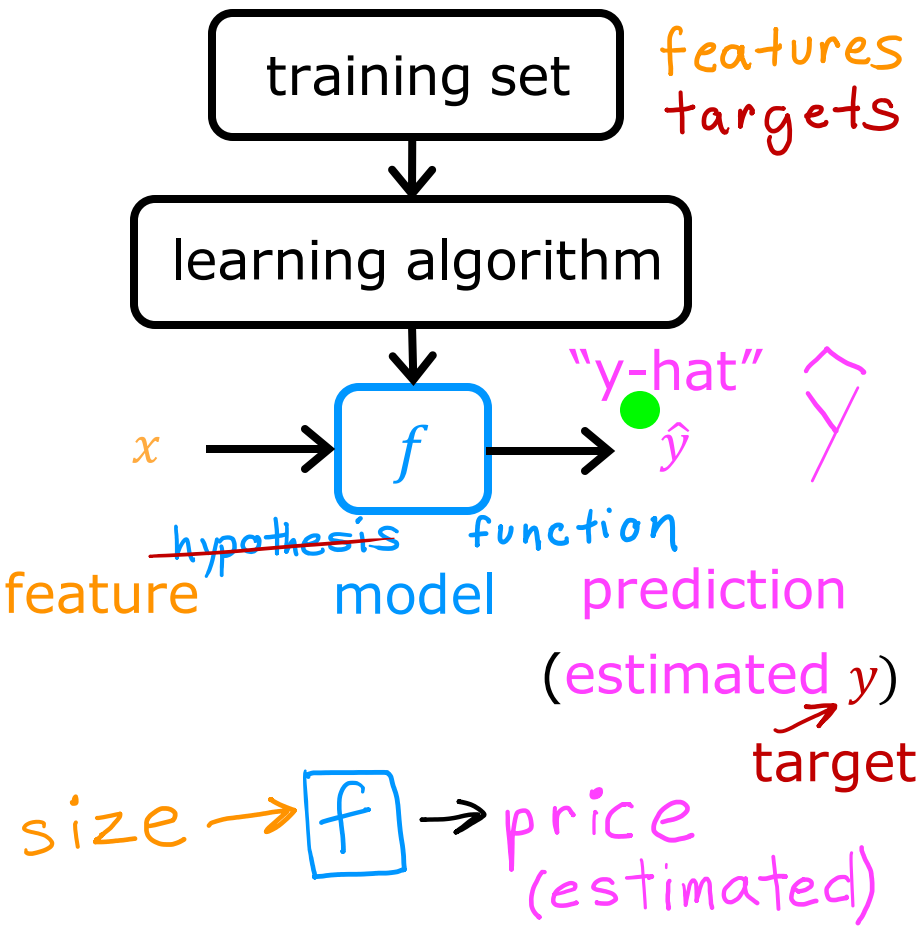
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Linear Regression with One Variable

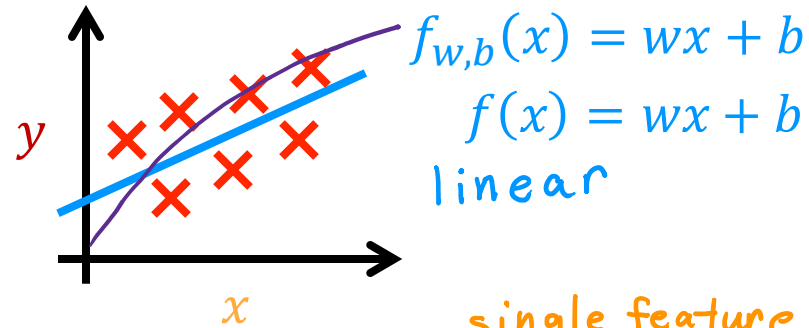
Linear Regression Model Part 2



How to represent f ?

$$f_{w,b}(x) = wx + b$$

$$f(x)$$



single feature x

Linear regression with one variable.

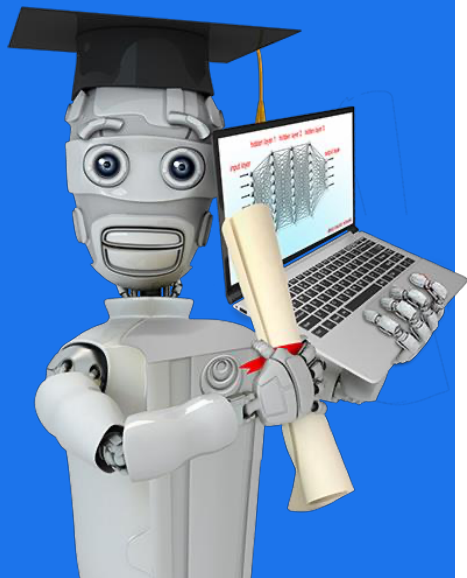
size

Univariate linear regression.

one variable

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Linear Regression with One Variable

Cost Function

Training set

^{features} size in feet ² (x)	^{targets} price \$1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

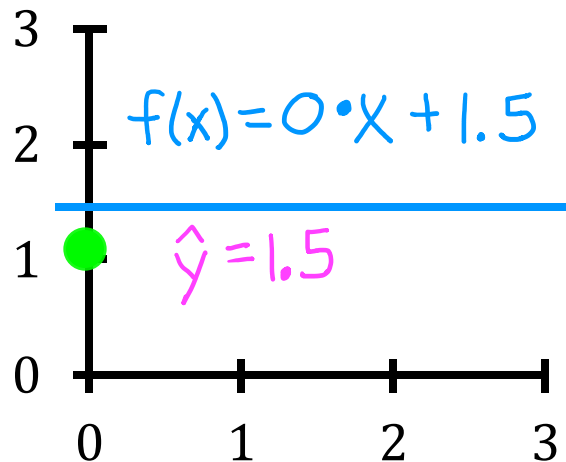
Model: $f_{w,b}(x) = wx + b$

w, b : parameters
coefficients
weights

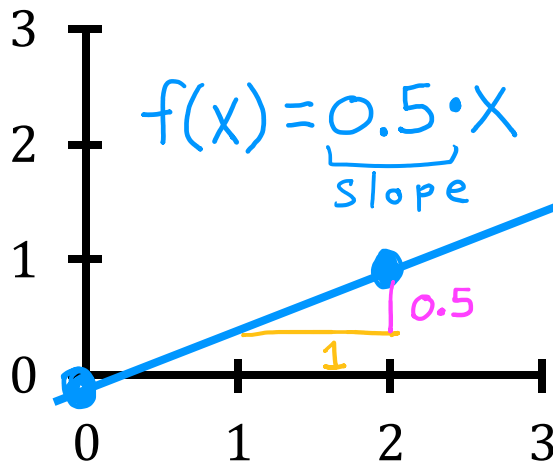
What do w, b do?

$$\underline{f_{w,b}}(x) = wx + b$$

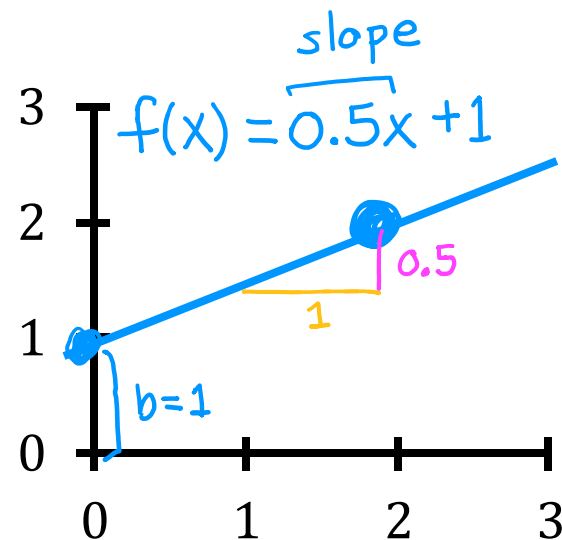
$f(x)$



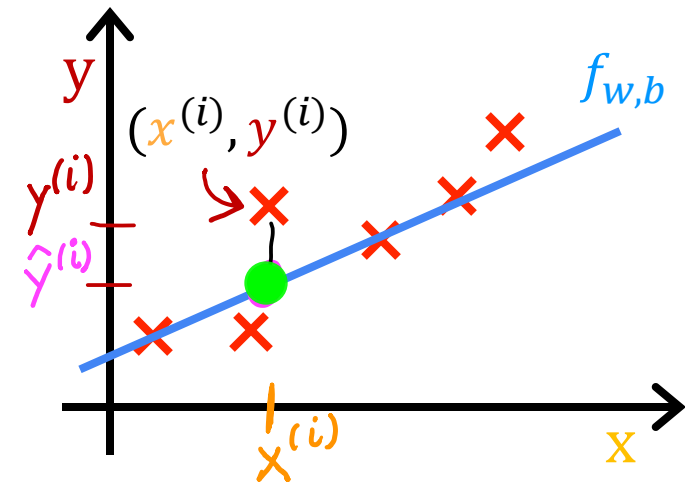
→ $w = 0$
 → $b = 1.5$
 (y-intercept)



→ $w = 0.5$
 → $b = 0$



→ $w = 0.5$
 → $b = 1$



$$\hat{y}^{(i)} = f_{w,b}(x^{(i)}) \quad \leftarrow$$

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

Cost function: Squared error cost function

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m \left(\hat{y}^{(i)} - y^{(i)} \right)^2$$

error

m = number of training examples

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m \left(f_{w,b}(x^{(i)}) - y^{(i)} \right)^2$$

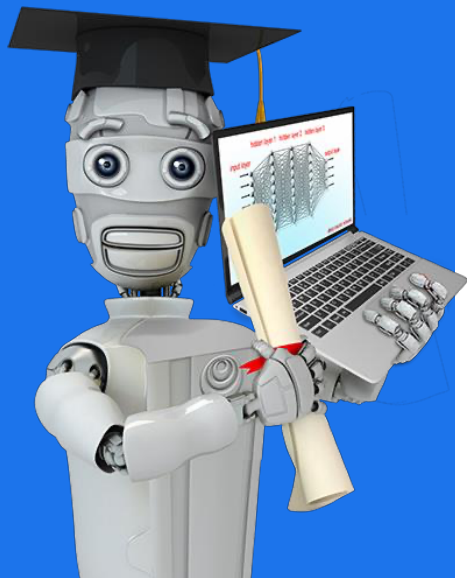
intuition (next!)

Find w, b :

$\hat{y}^{(i)}$ is close to $y^{(i)}$ for all $(x^{(i)}, y^{(i)})$.

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Linear Regression with One Variable

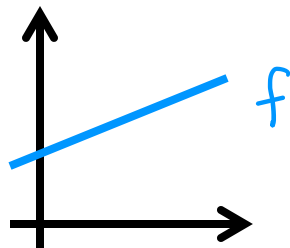
Cost Function Intuition

model:

$$\underline{f_{w,b}(x) = wx + b}$$

parameters:

$$\underline{w, b}$$



cost function:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

goal:

$$\underset{w, b}{\text{minimize}} J(w, b)$$

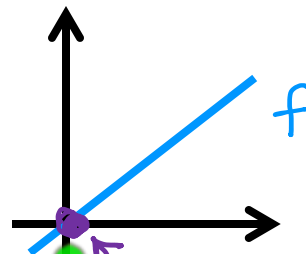
simplified

$$f_w(x) = \underline{wx}$$

$$b = \emptyset$$

w

w



$$\underline{J(w)} = \frac{1}{2m} \sum_{i=1}^m (\underline{f_w(x^{(i)})} - y^{(i)})^2$$

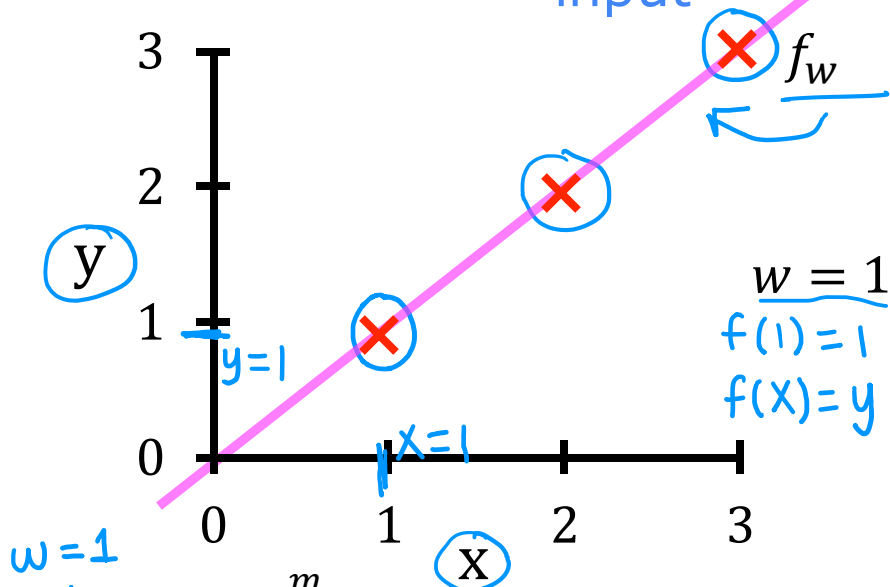
$$wx^{(i)}$$

$$\underset{\underline{w}}{\text{minimize}} \underline{J(w)}$$

→ $f_w(x)$

(for fixed w , function of x)

input

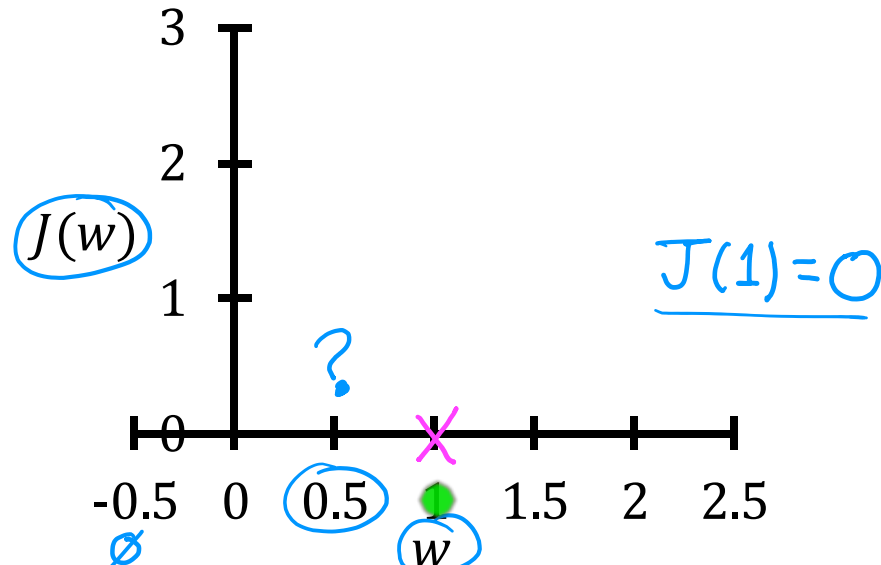


$$J(w) = \frac{1}{2m} \sum_{i=1}^m (f_w(x^{(i)}) - y^{(i)})^2 = \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} - y^{(i)})^2$$

$J(w)$

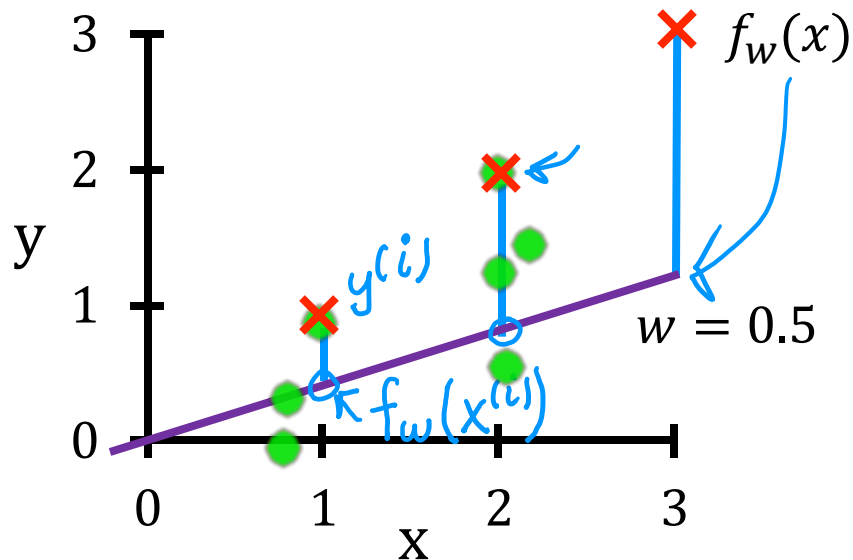
(function of w)

parameter



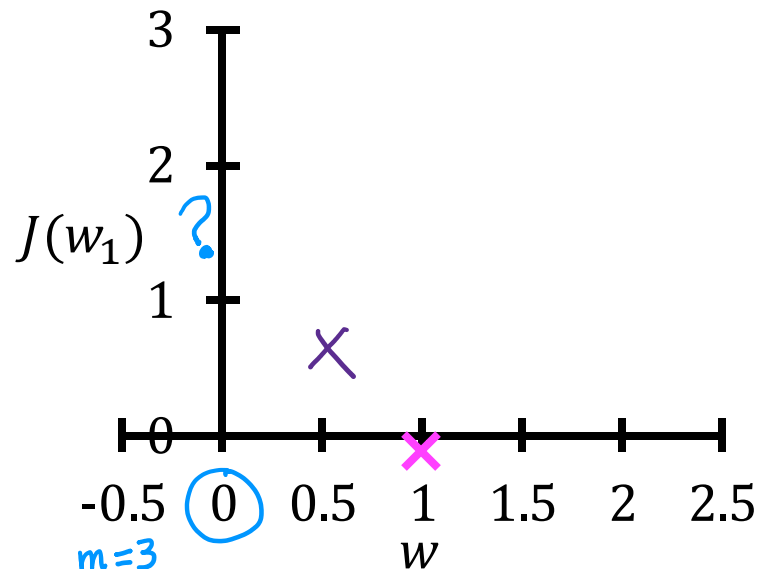
$$= \frac{1}{2m} (0^2 + 0^2 + 0^2) = 0$$

$f_w(x)$
(function of x)

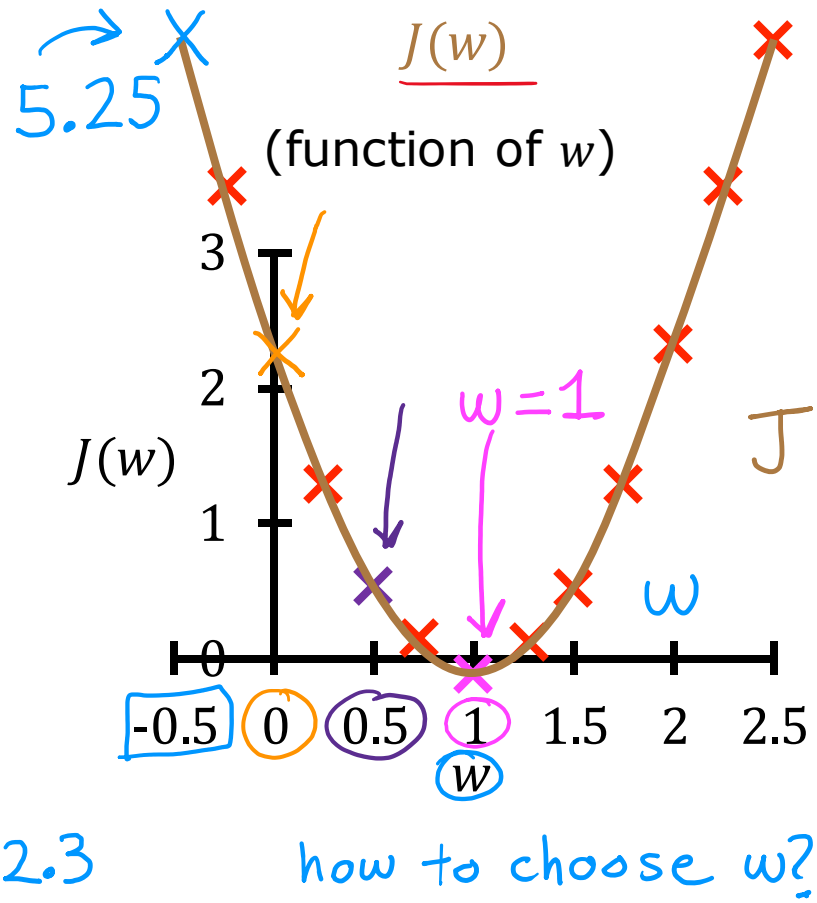
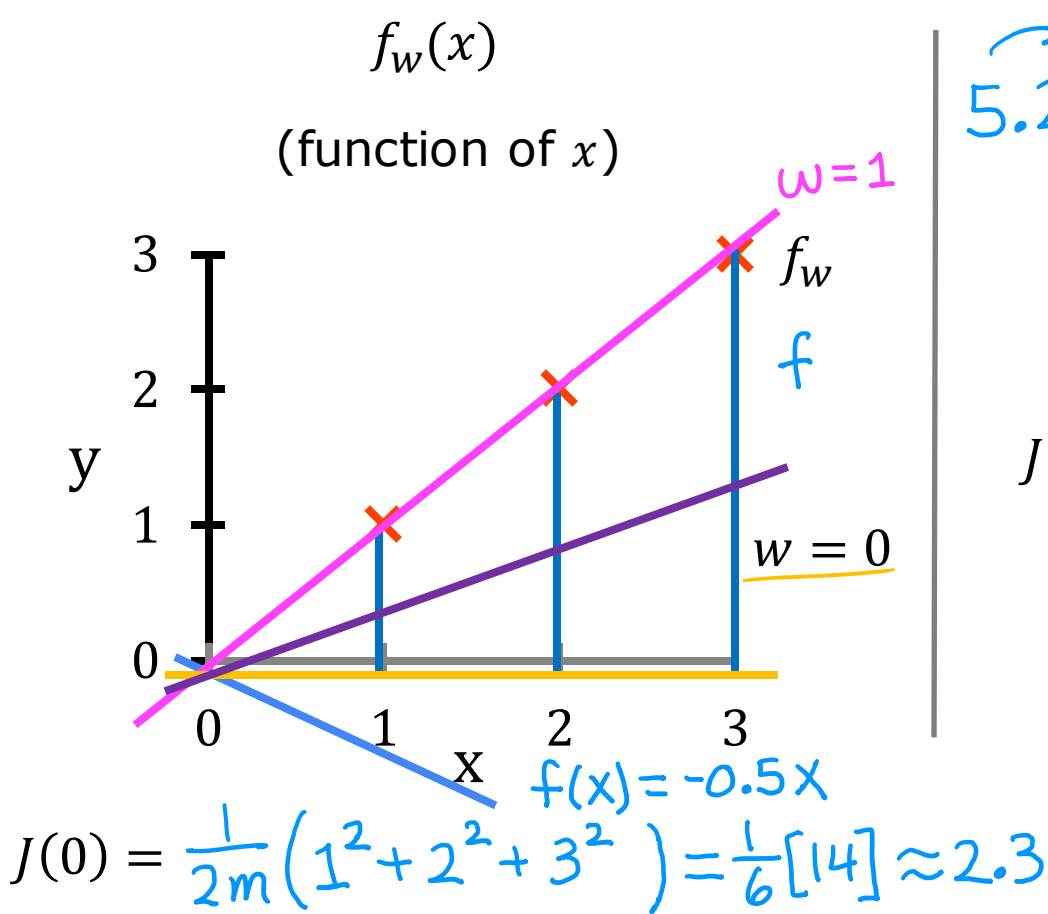


$J(w)$

(function of w)



$$J(0.5) = \frac{1}{2m} [(0.5-1)^2 + (1-2)^2 + (1.5-3)^2] = \frac{1}{2 \times 3} [3.5] = \frac{3.5}{6} \approx 0.58$$

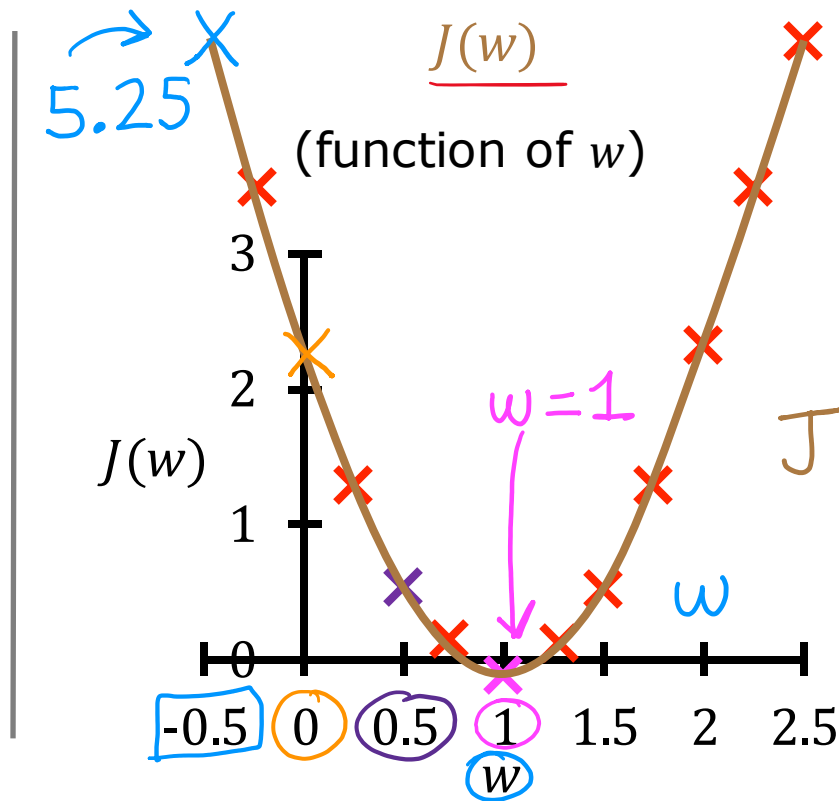


goal of linear regression:

minimize $J(w)$

general case:

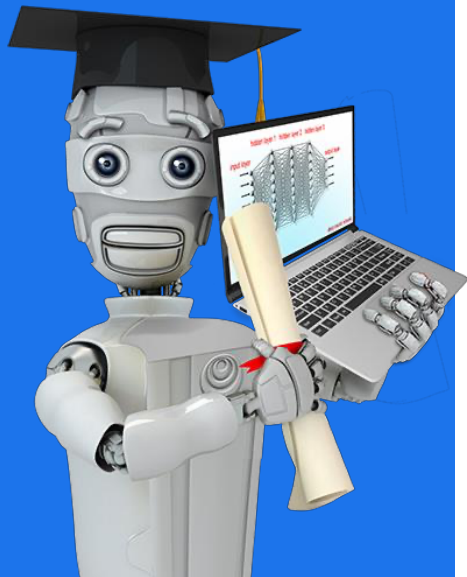
minimize $J(w, b)$



choose w to minimize $J(w)$

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Linear Regression with One Variable

Visualizing
the Cost Function

Model $f_{w,b}(x) = wx + b$

Parameters

w, b

~~before: $b=0$~~

Cost Function

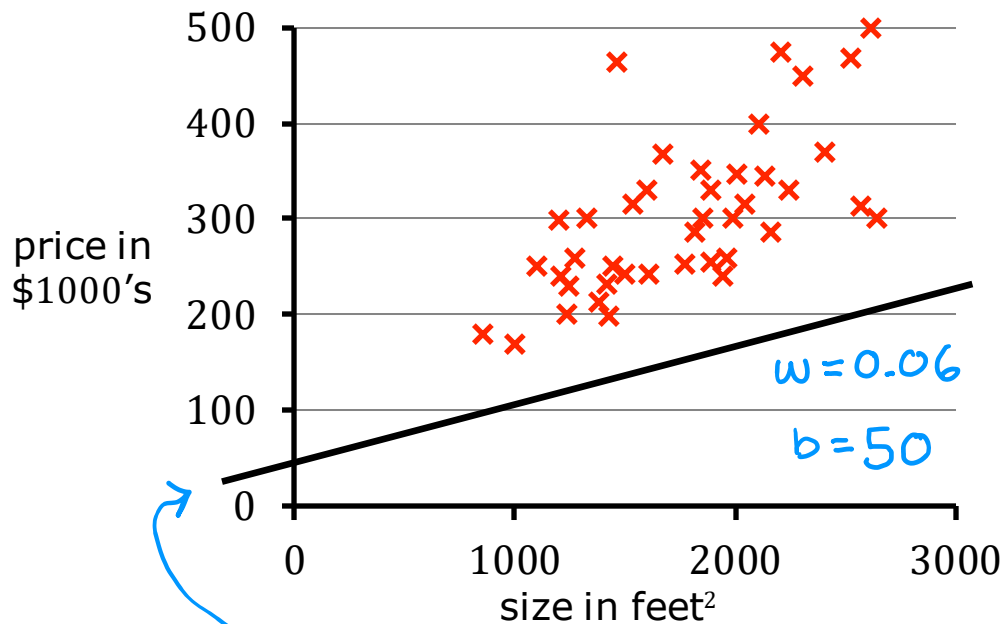
$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Objective

minimize $J(w, b)$
 w, b

$f_{w,b}$

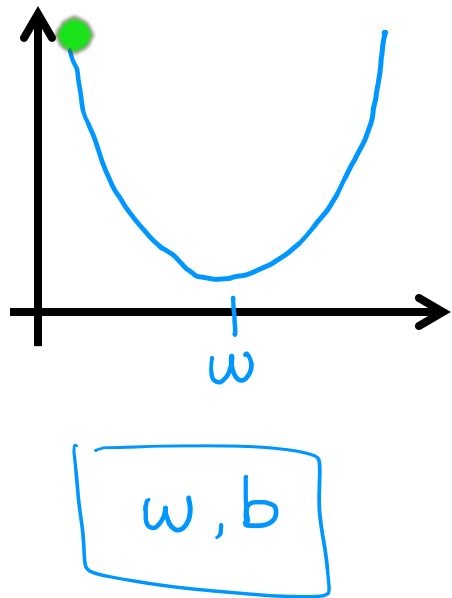
(function of x)

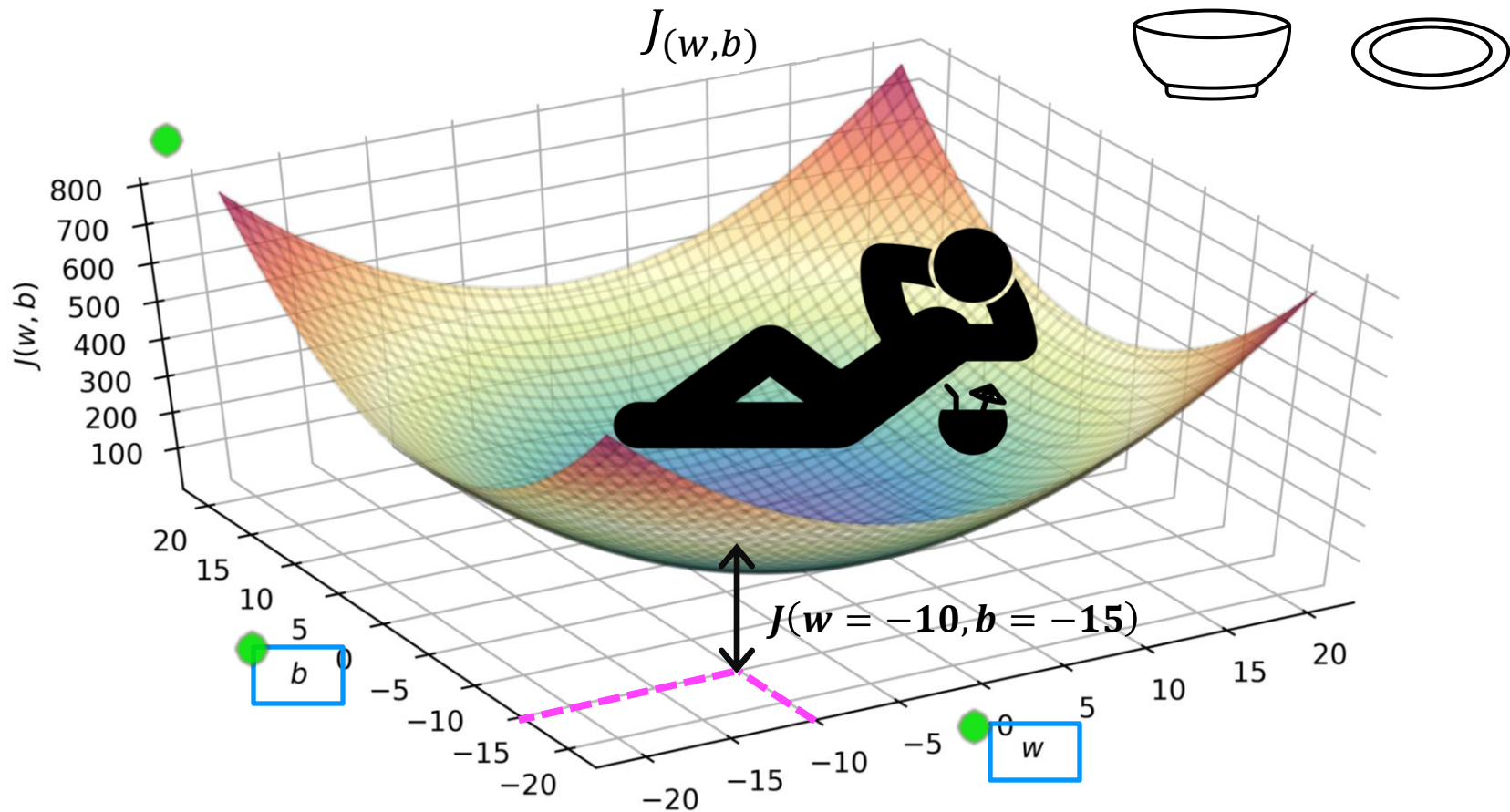


$$f_{w,b}(x) = 0.06x + 50$$

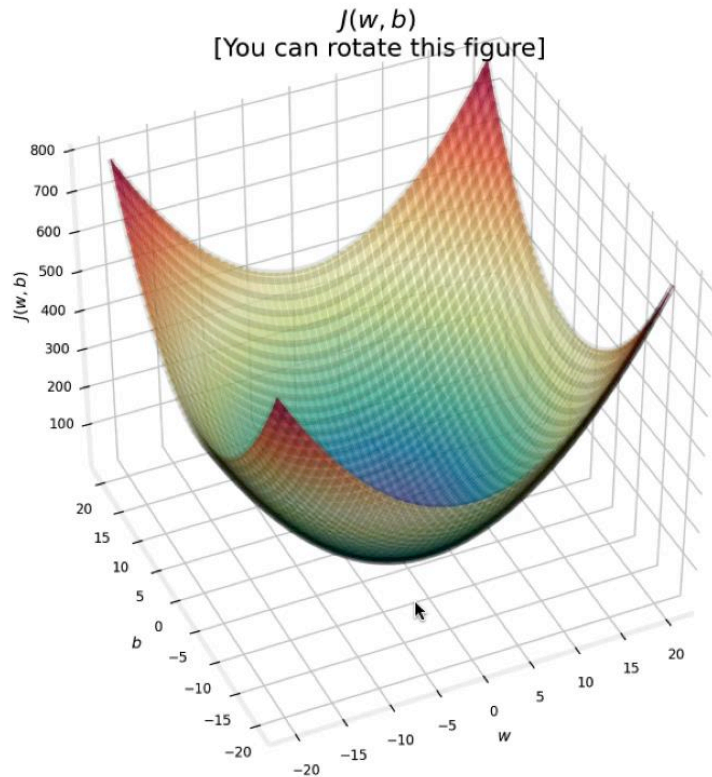
J

(function of w, b)



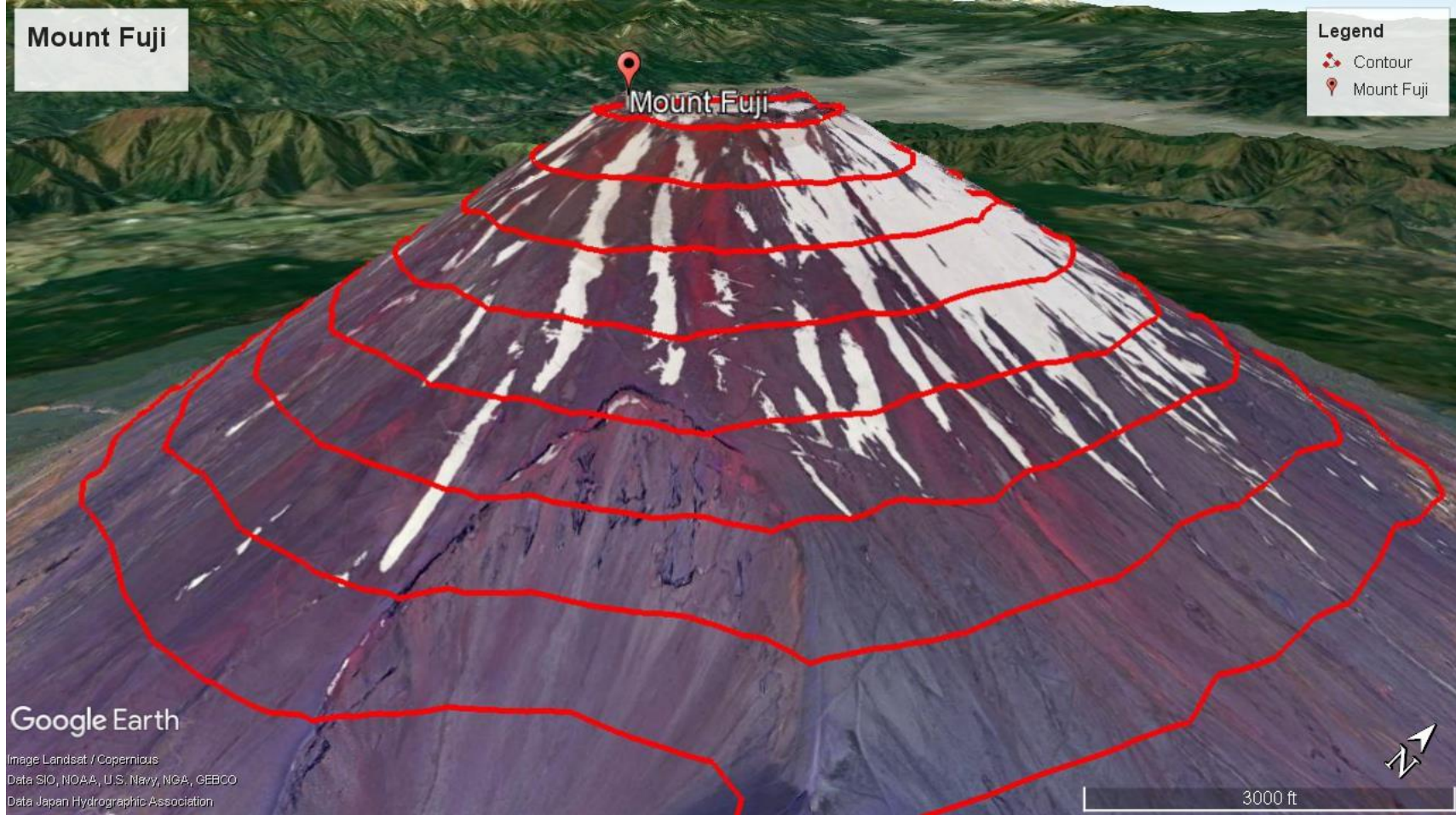


3D surface plot

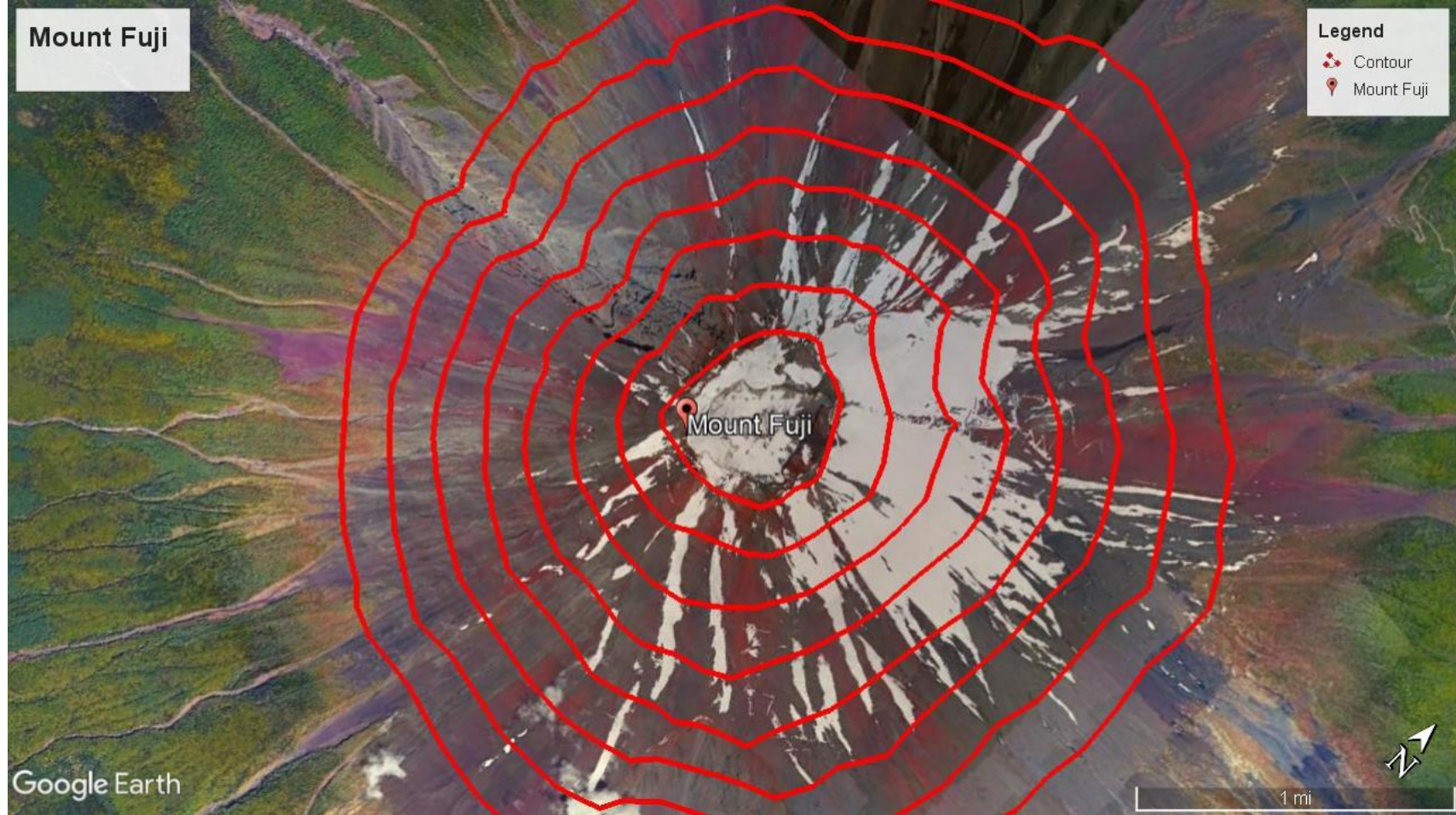


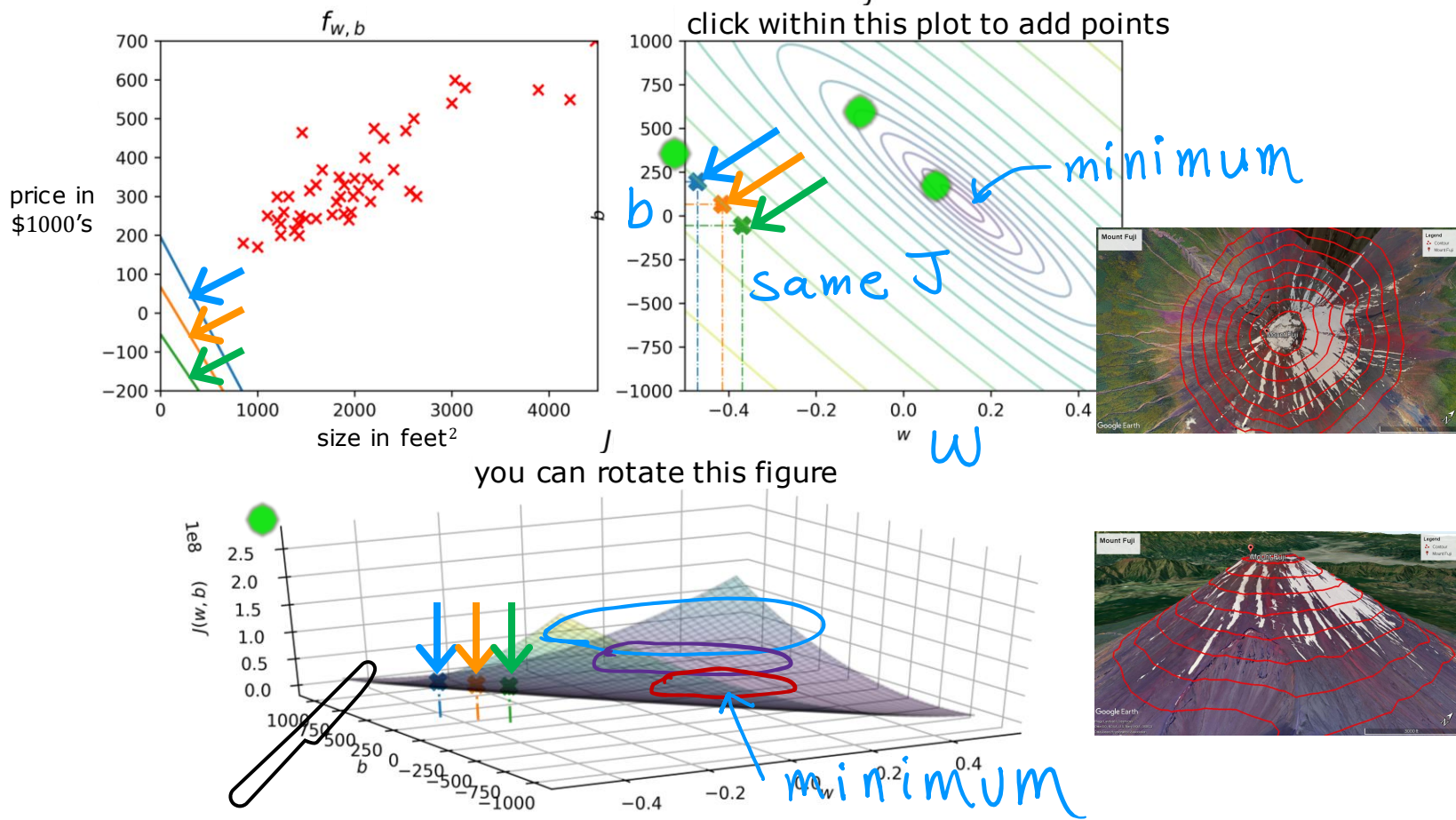
Alternative
contour plot

Mount Fuji



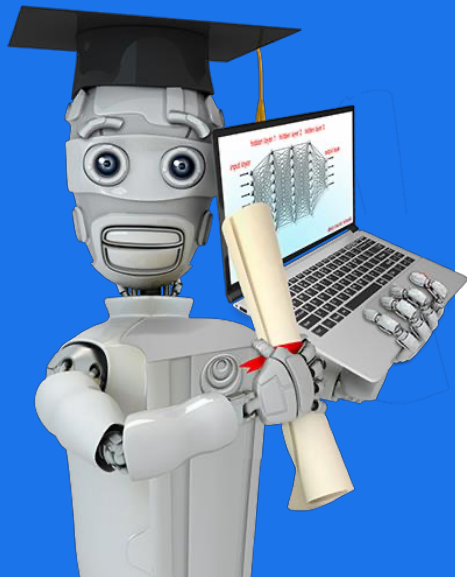
Mount Fuji





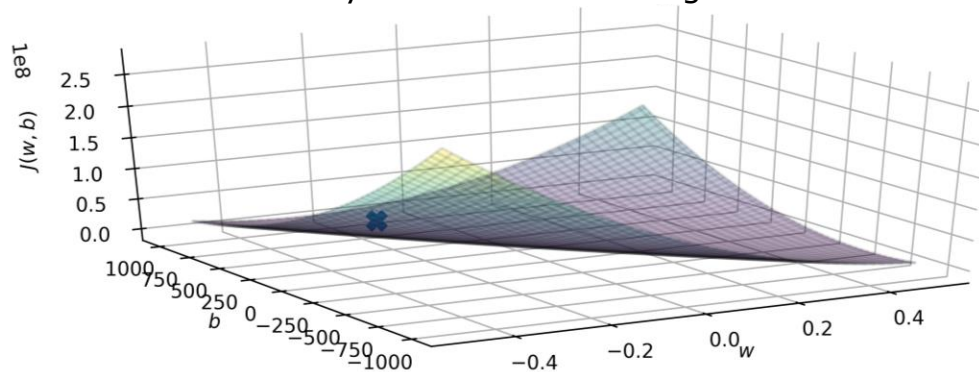
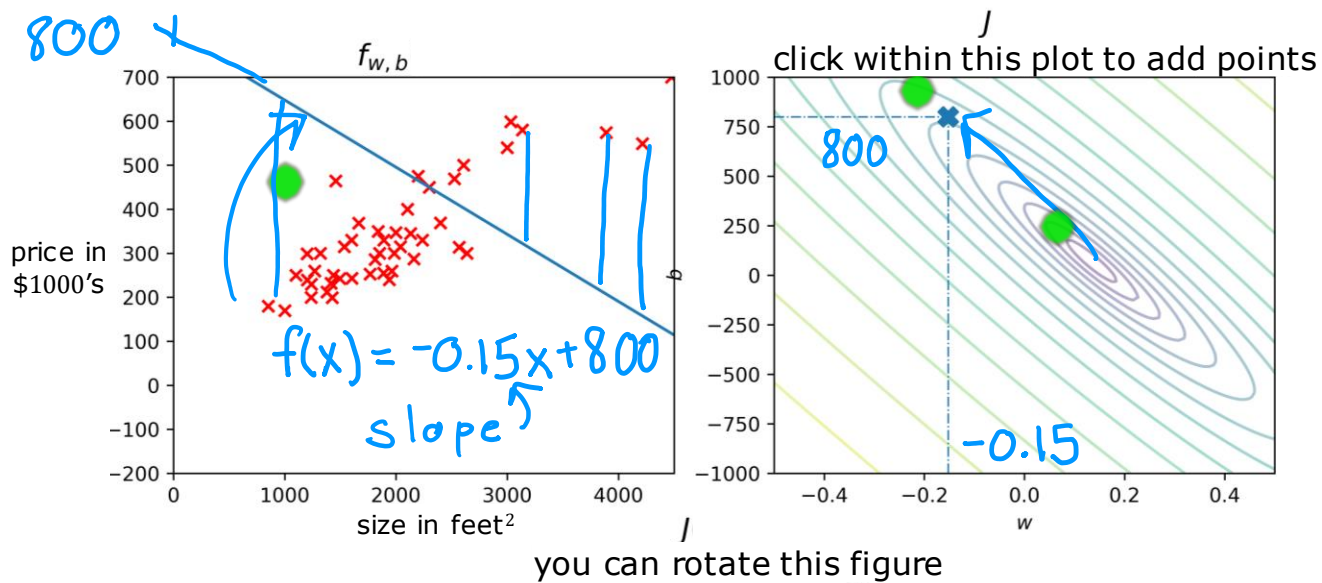
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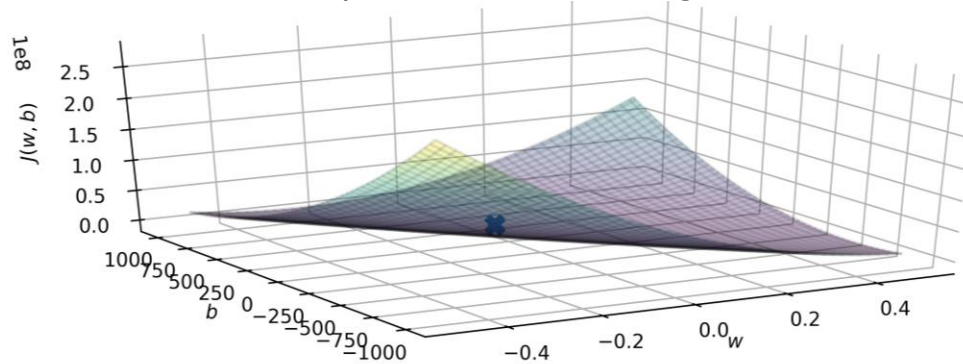
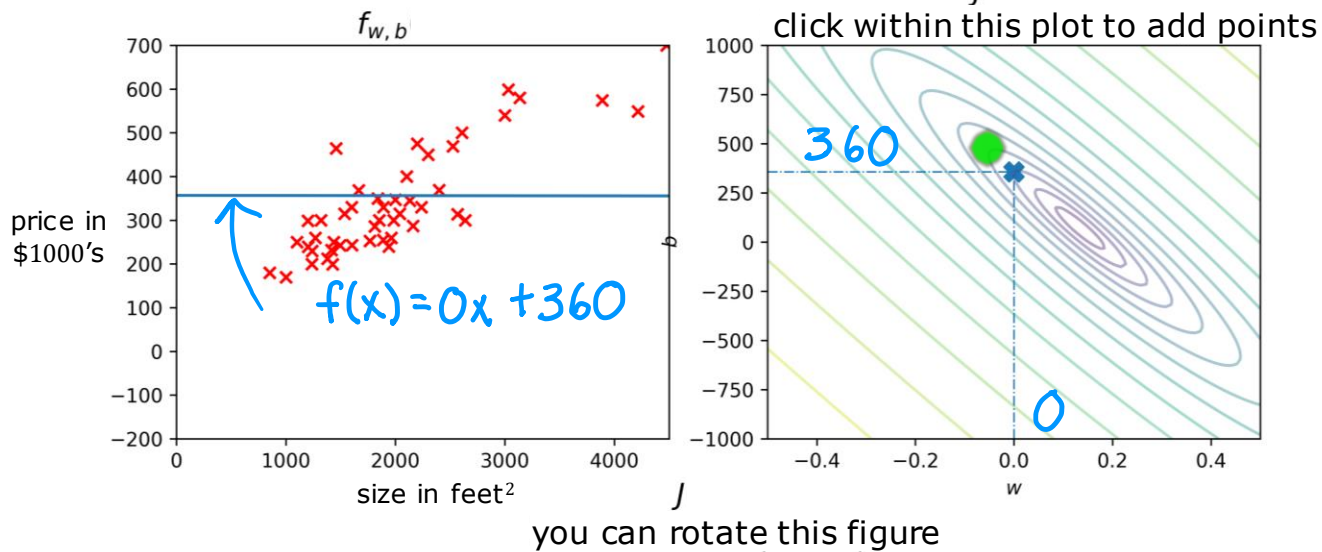
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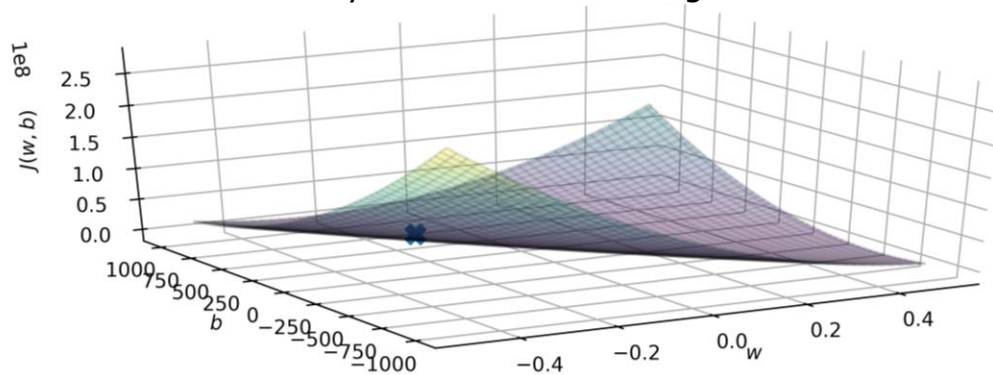
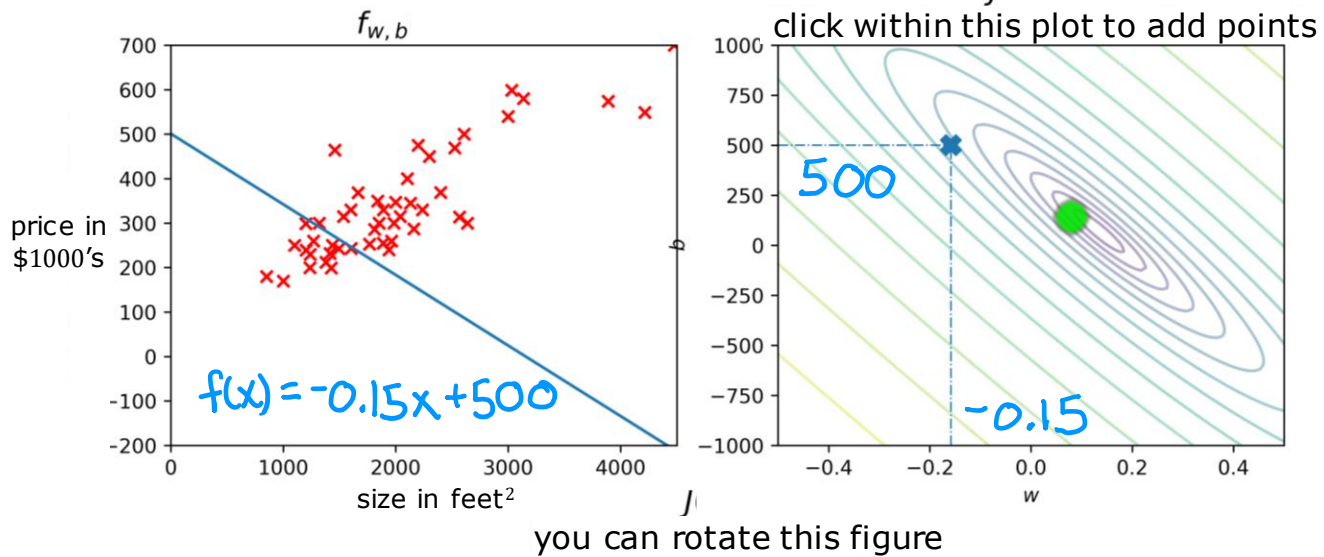


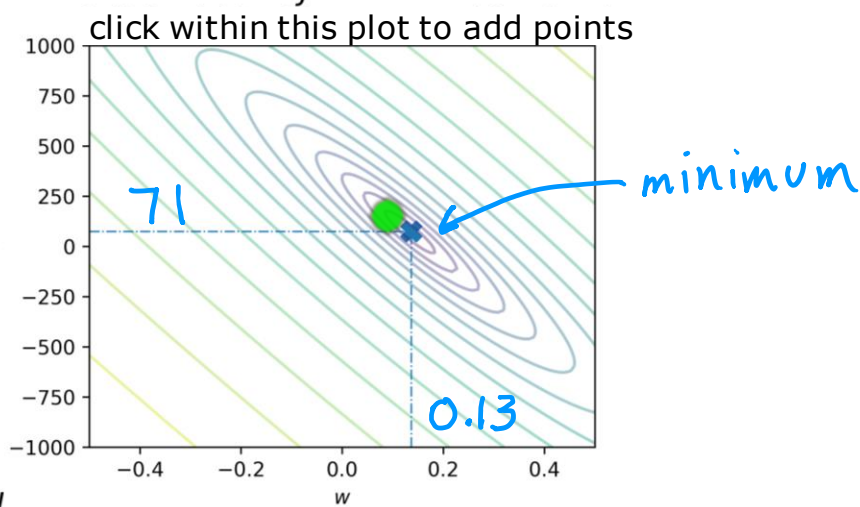
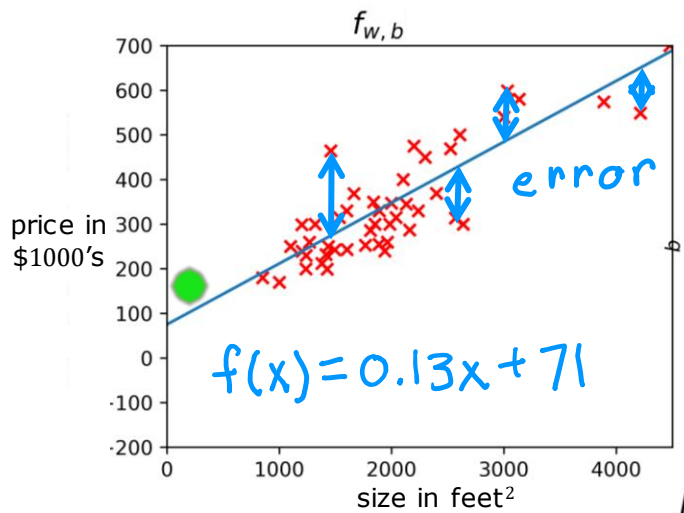
Linear Regression with One Variable

Visualization examples

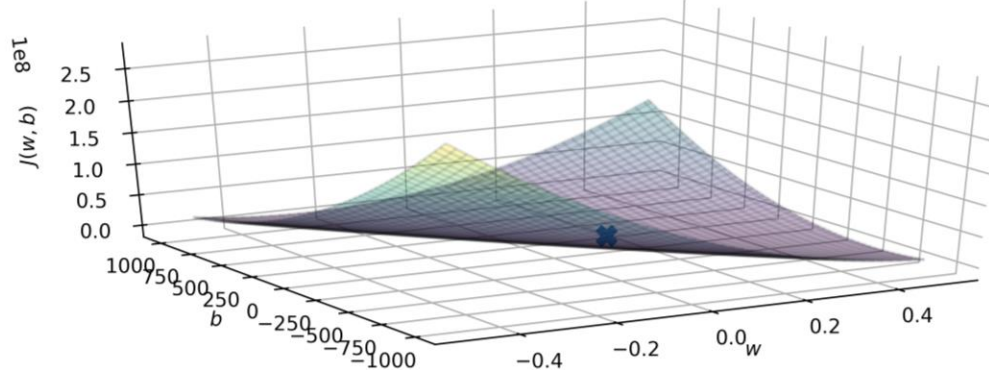






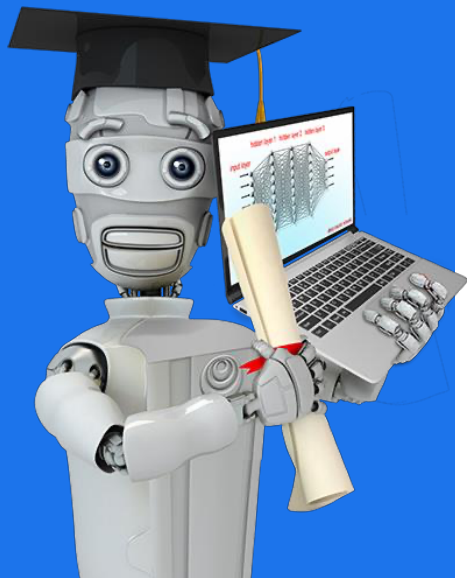


you can rotate this figure



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Training Linear Regression

Gradient Descent

Have some function $J(w, b)$ *for linear regression
or any function*

Want $\min_{w, b} J(w, b)$ $\min_{w_1, \dots, w_n, b} \underline{J(w_1, w_2, \dots, w_n, b)}$

Outline:

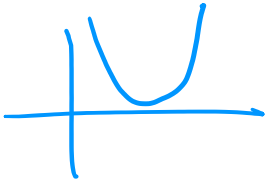
Start with some w, b (*set $w=0, b=0$*)

Keep changing w, b to reduce $J(w, b)$

Until we settle at or near a minimum

may have >1 minimum

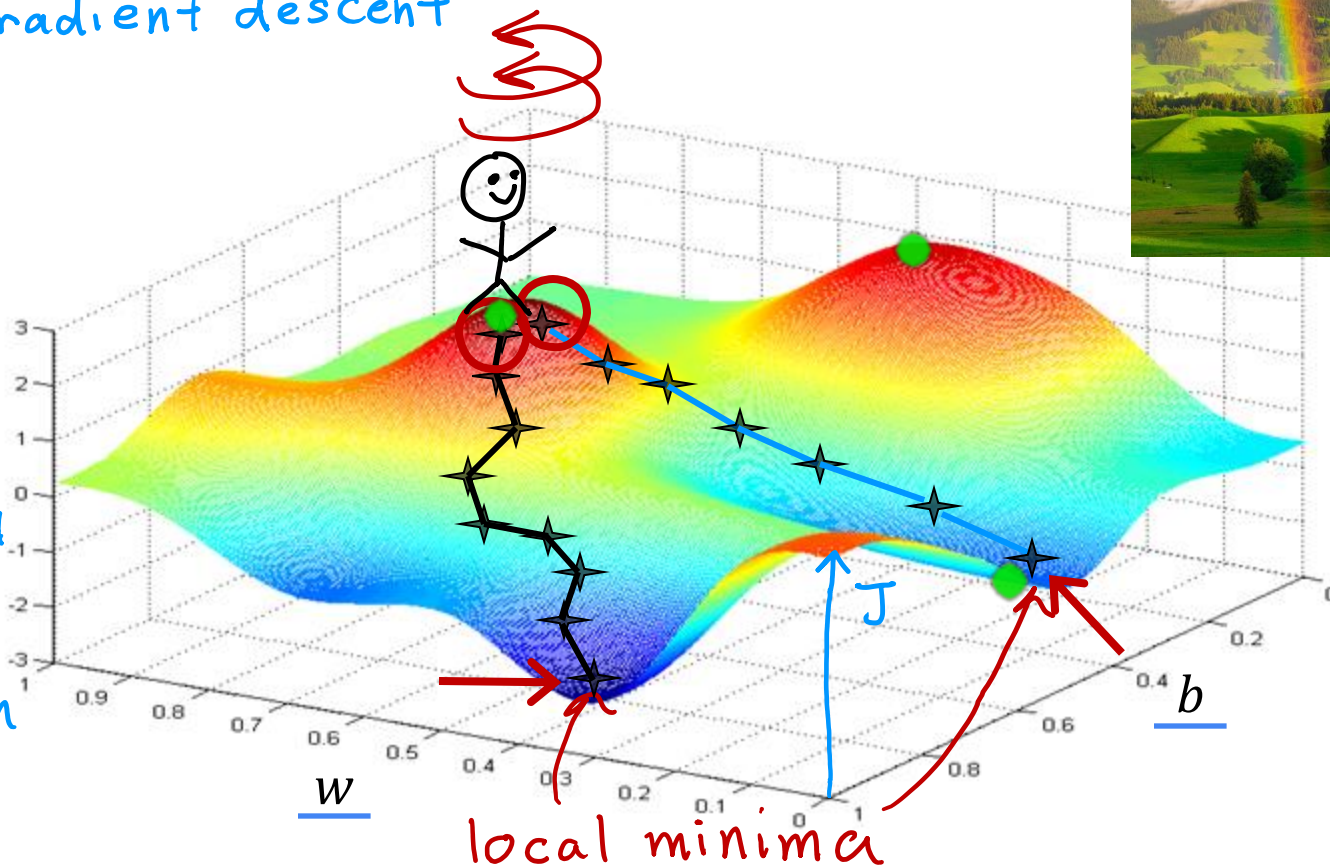
J not always



gradient descent

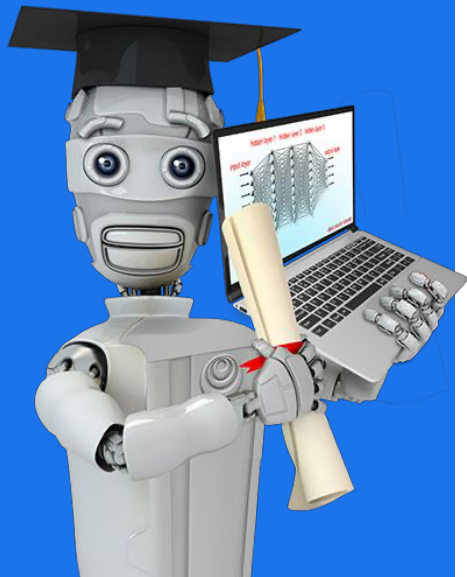


$J(w, b)$
not squared
error cost
not linear
regression



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Training Linear Regression

Implementing Gradient Descent

Gradient descent algorithm

Repeat until convergence

$$\begin{cases} \underline{w} = w - \alpha \frac{\partial}{\partial w} J(w, b) \\ \underline{b} = b - \alpha \frac{\partial}{\partial b} J(w, b) \end{cases}$$

Learning rate
Derivative

Simultaneously
update w and b

Assignment

$$\begin{aligned} a &= c \\ a &= a + 1 \end{aligned}$$

Code

Truth assertion

$$\begin{aligned} a &= c \\ a &= a + 1 \end{aligned}$$

Math
 $a == c$

Correct: Simultaneous update

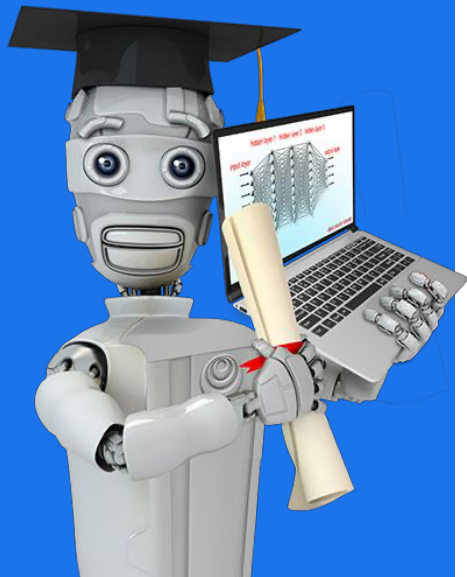
$$\begin{aligned} tmp_w &= w - \alpha \frac{\partial}{\partial w} J(w, b) \\ tmp_b &= b - \alpha \frac{\partial}{\partial b} J(w, b) \\ w &= tmp_w \\ b &= tmp_b \end{aligned}$$

Incorrect

$$\begin{aligned} tmp_w &= w - \alpha \frac{\partial}{\partial w} J(w, b) \\ w &= tmp_w \\ tmp_b &= b - \alpha \frac{\partial}{\partial b} J(w, b) \\ b &= tmp_b \end{aligned}$$

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Training Linear Regression

Gradient Descent Intuition

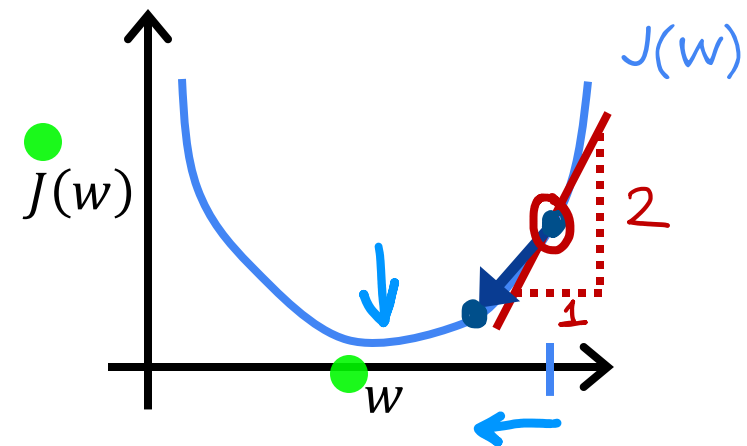
Gradient descent algorithm

● repeat until convergence {

learning rate α derivative

$$\begin{cases} \underline{w} = w - \alpha \frac{\partial}{\partial w} J(w, b) \\ \underline{b} = b - \alpha \frac{\partial}{\partial b} J(w, b) \end{cases}$$

$$J(w)$$
$$w = w - \alpha \frac{\partial}{\partial w} J(w)$$
$$\underline{\min}_w J(w)$$



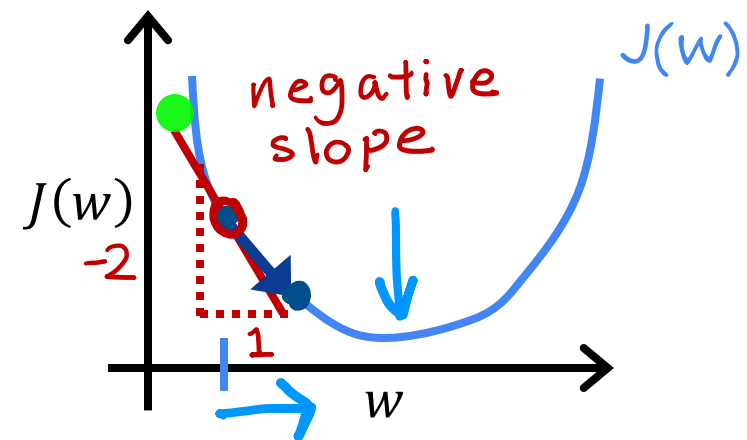
$$w = w - \alpha \underbrace{\frac{d}{dw} J(w)}_{> 0}$$

$$w = w - \underline{\alpha} \cdot (\text{positive number})$$

$$\underline{\frac{d}{dw} J(w)} < 0$$

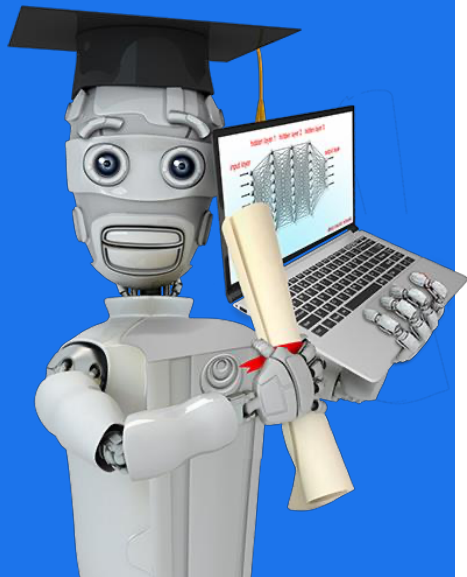
$$w = \underline{w} - \alpha \cdot (\text{negative number})$$

↑ ↑



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Training Linear Regression

Learning Rate

$$w = w - \alpha \frac{d}{dw} J(w)$$

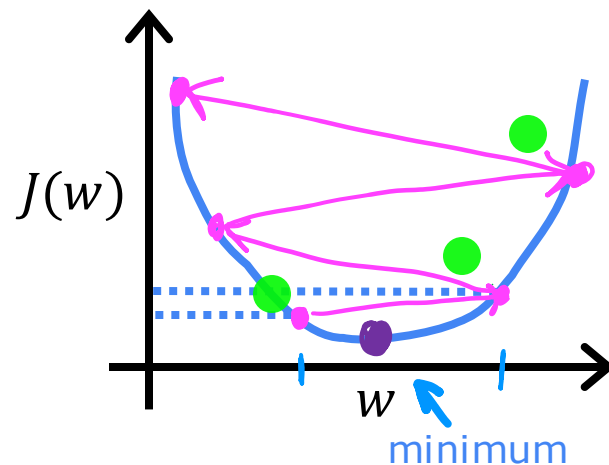
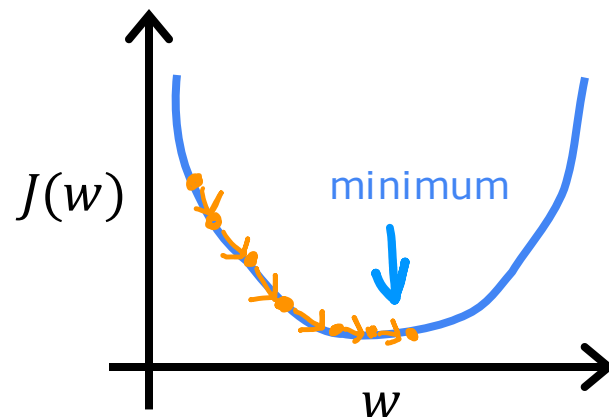
If α is too small...

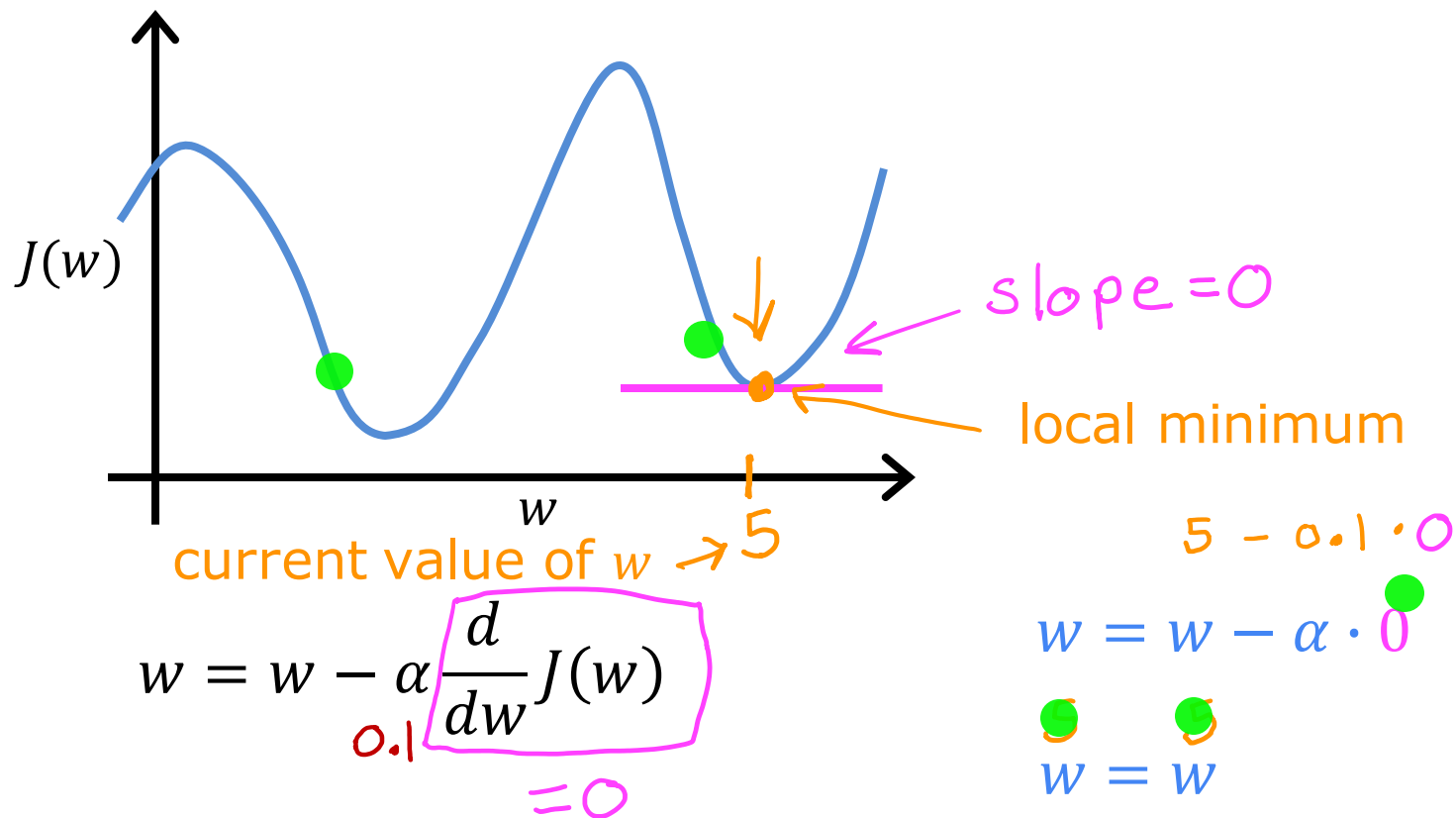
Gradient descent may be slow.

If α is too large...

Gradient descent may:

- Overshoot, never reach minimum
- Fail to converge, diverge





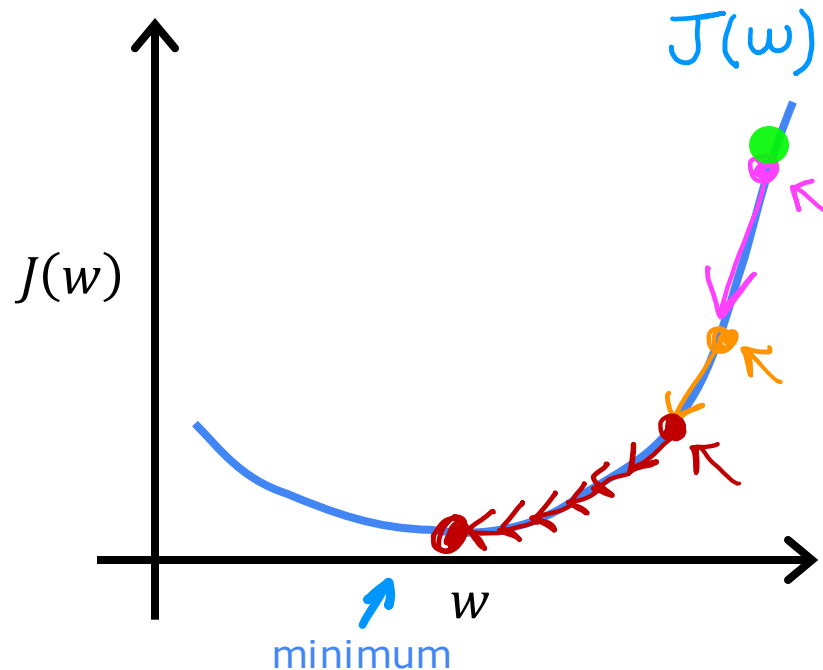
Can reach local minimum with fixed learning rate α

$$w = w - \underbrace{\alpha}_{\text{smaller}} \underbrace{\frac{d}{dw} J(w)}_{\text{not as large}} \underbrace{J(w)}_{\text{large}}$$

Near a local minimum,

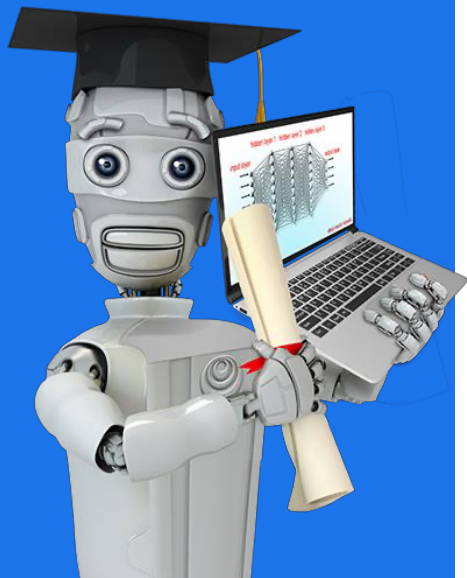
- Derivative becomes smaller
- Update steps become smaller

Can reach minimum without decreasing learning rate α



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Training Linear Regression

Gradient Descent for Linear Regression

Linear regression model

$$f_{w,b}(x) = wx + b$$

Cost function

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Gradient descent algorithm

repeat until convergence {

$$w = w - \alpha \frac{\partial}{\partial w} J(w, b) \rightarrow \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})x^{(i)}$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w, b) \rightarrow \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

}

next slide
is optional!

(Optional)

$$\frac{\partial}{\partial w} J(w, b) = \frac{\partial}{\partial w} \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2 = \frac{\partial}{\partial w} \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)})^2$$

$$= \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)}) \cancel{2} x^{(i)} = \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$\frac{\partial}{\partial b} J(w, b) = \frac{\partial}{\partial b} \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2 = \frac{\partial}{\partial b} \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)})^2$$

$$= \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)}) \cancel{2} = \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

no $x^{(i)}$

Gradient descent algorithm

$$\frac{\partial}{\partial w} J(w, b)$$

repeat until convergence {

$$w = w - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

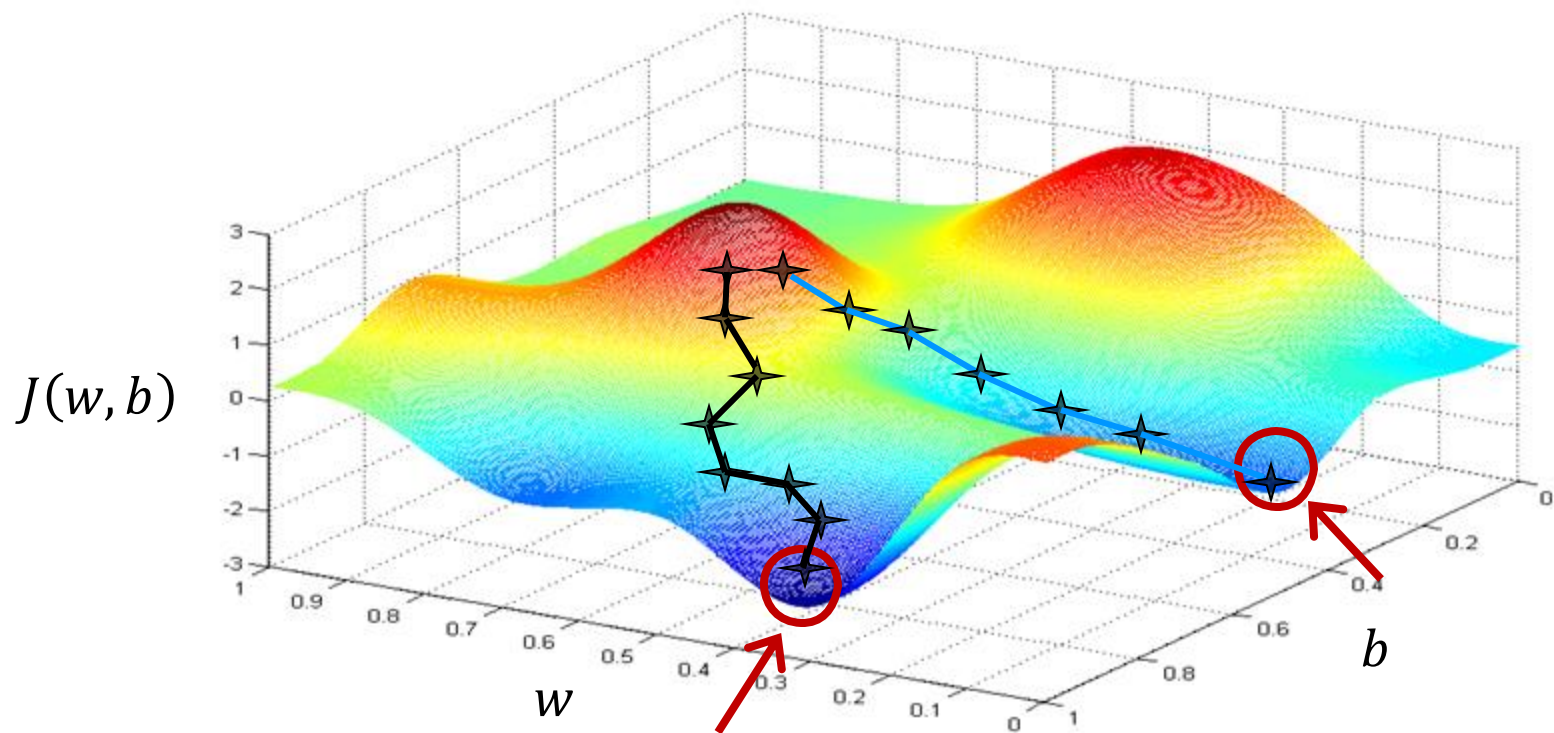
}

$$\frac{\partial}{\partial b} J(w, b)$$

Update
w and b
simultaneously

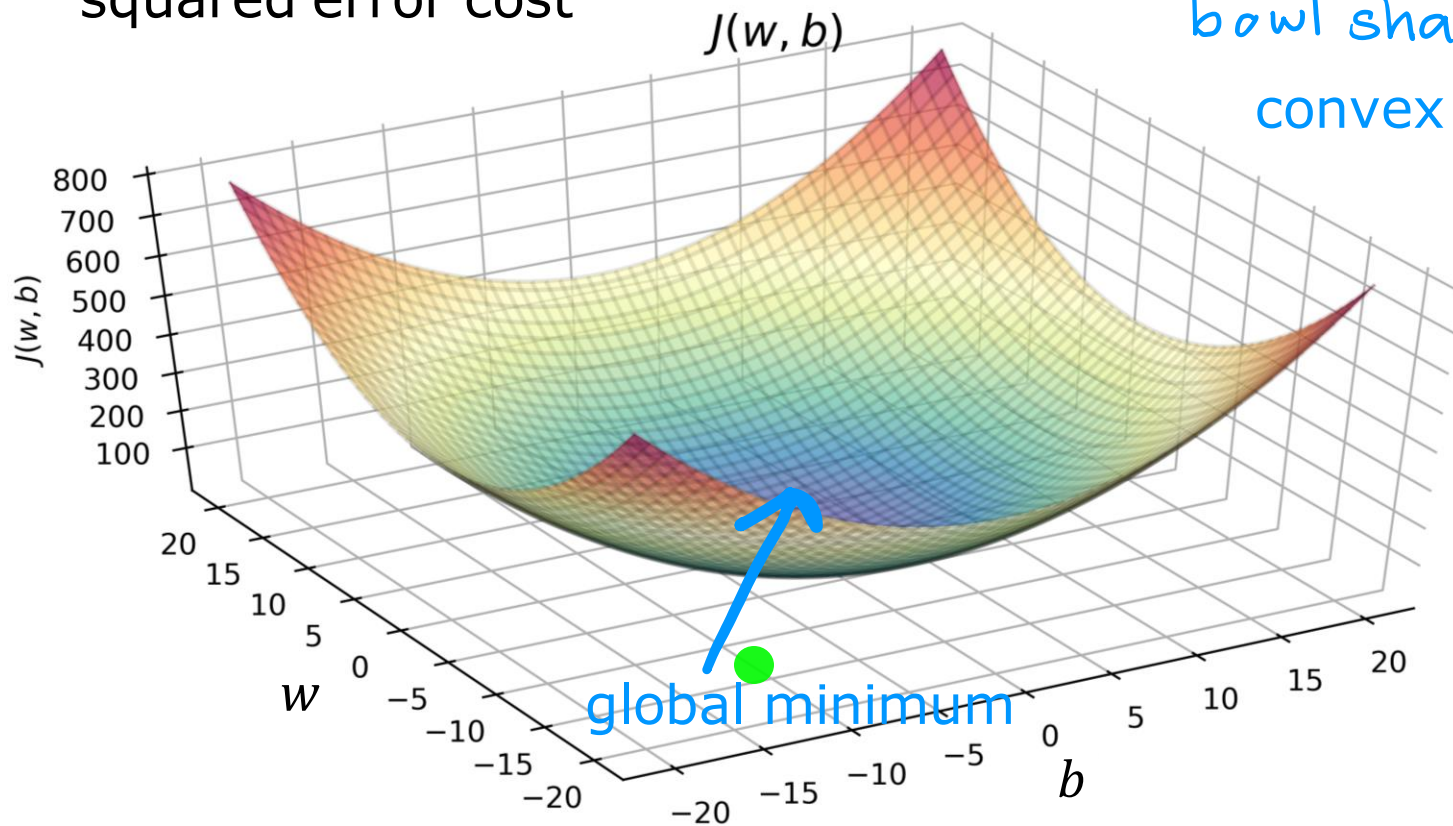
$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

More than one local minimum



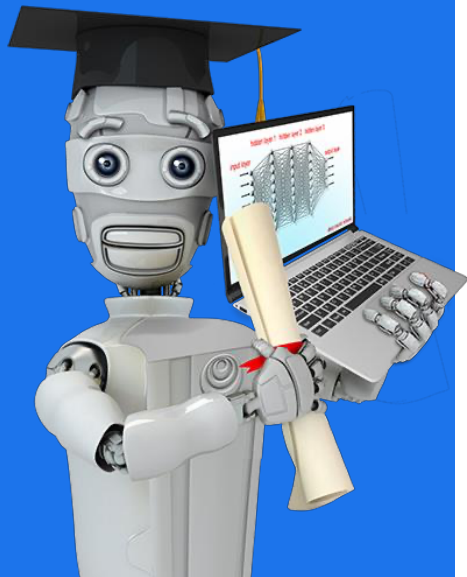
squared error cost

bowl shape 
convex function



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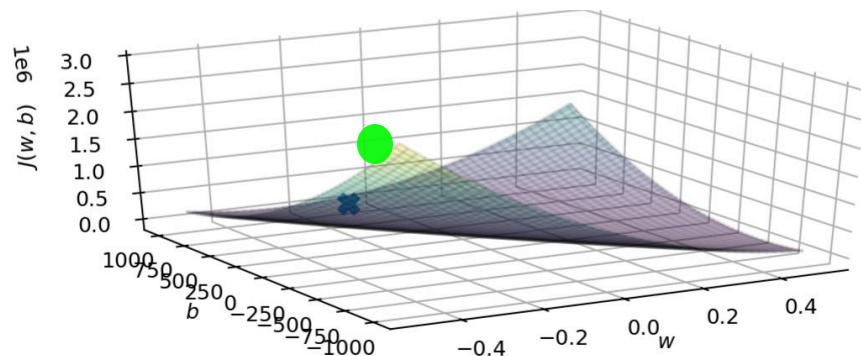
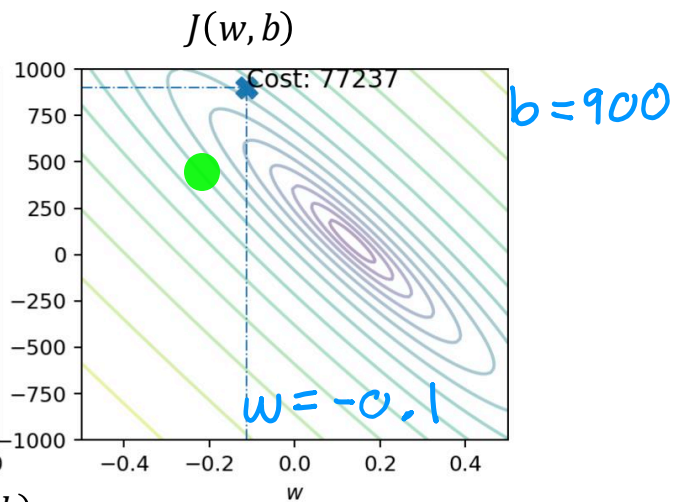
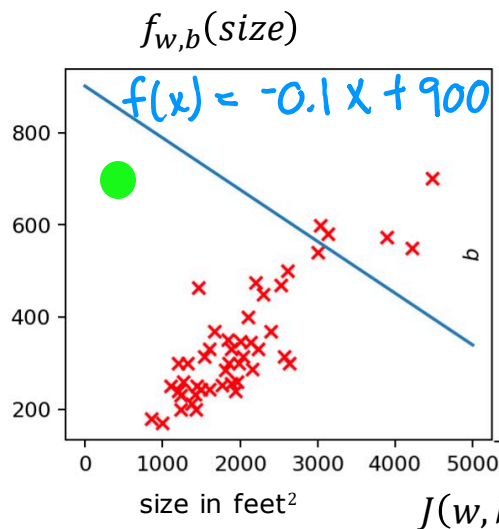
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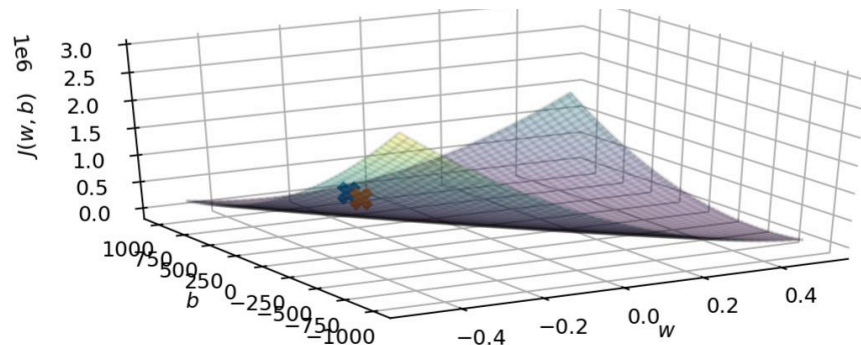
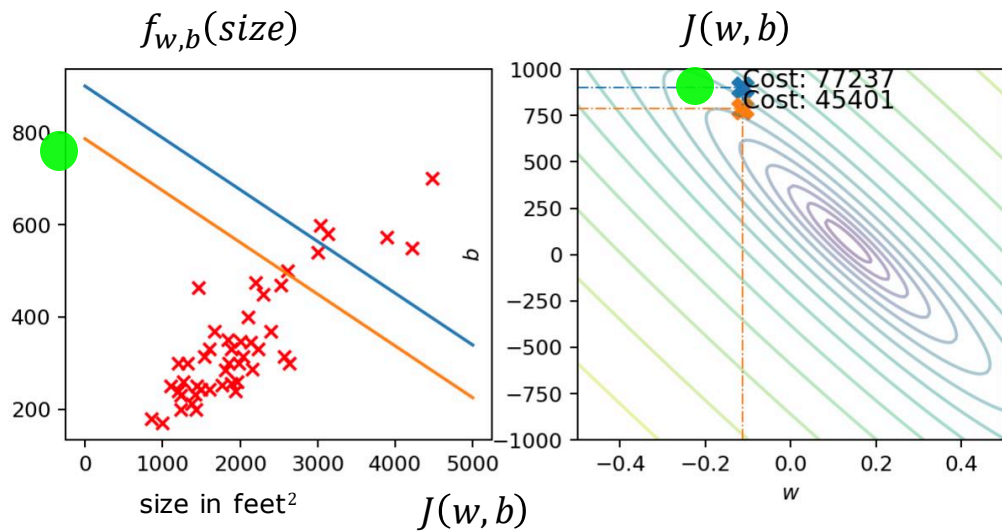
Training Linear Regression

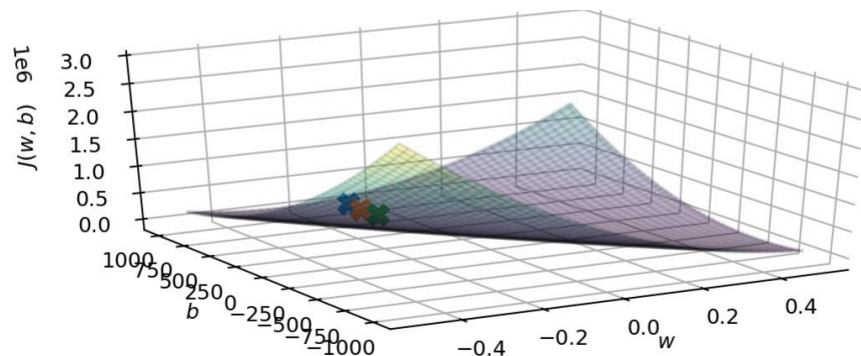
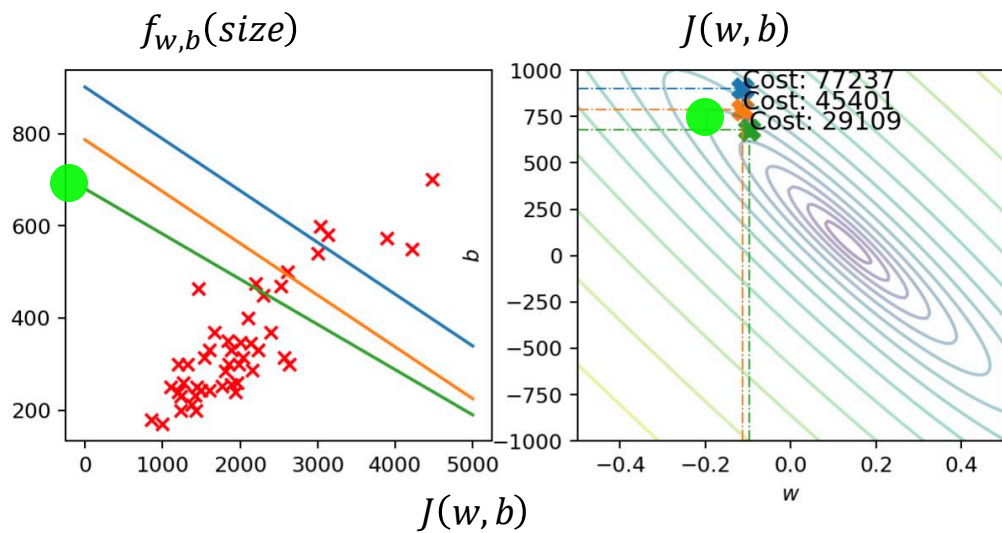
Running Gradient Descent

price in
\$1000's

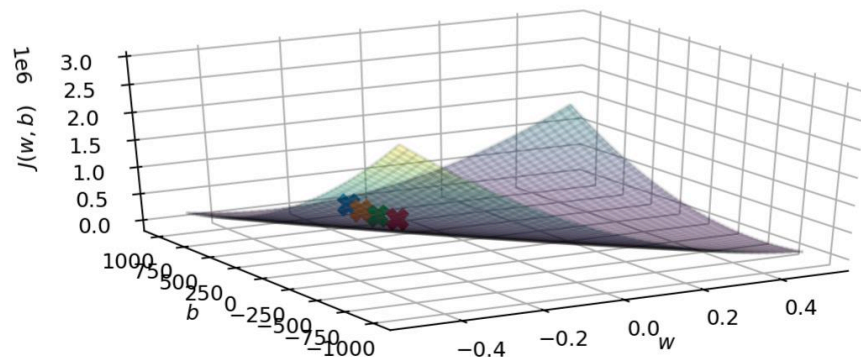
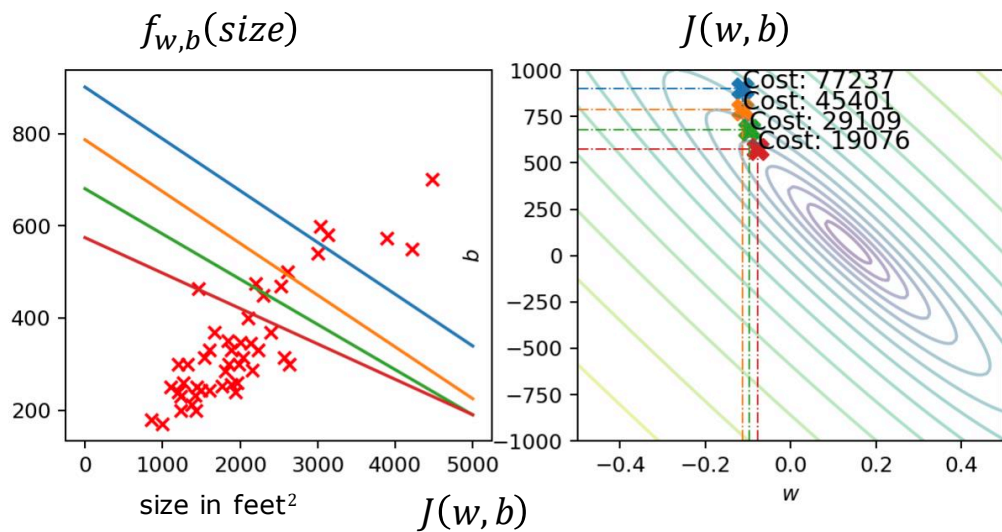


price in
\$1000's

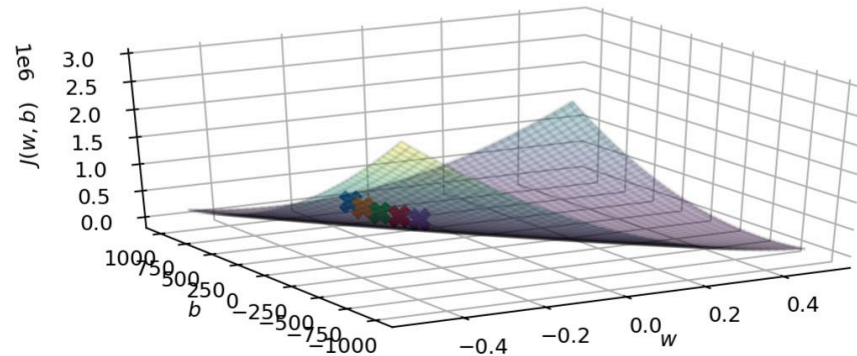
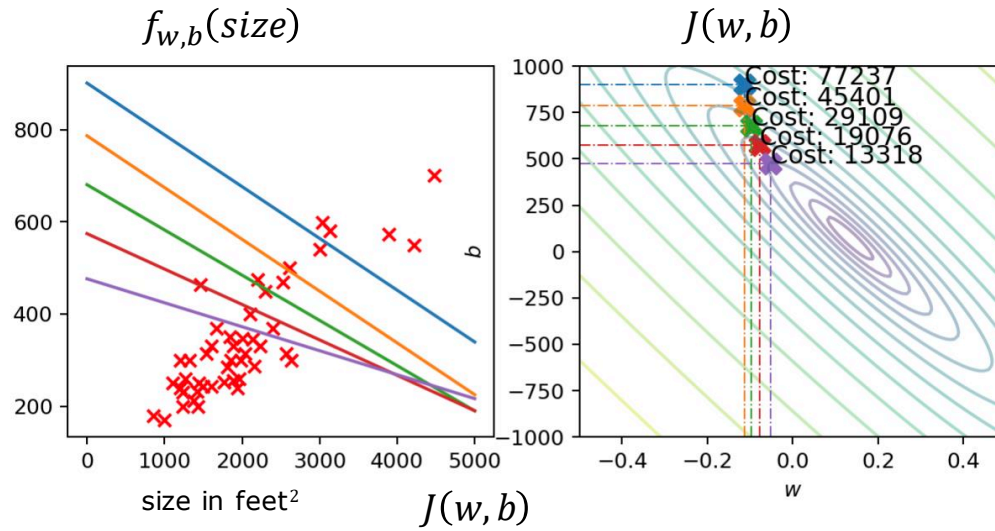




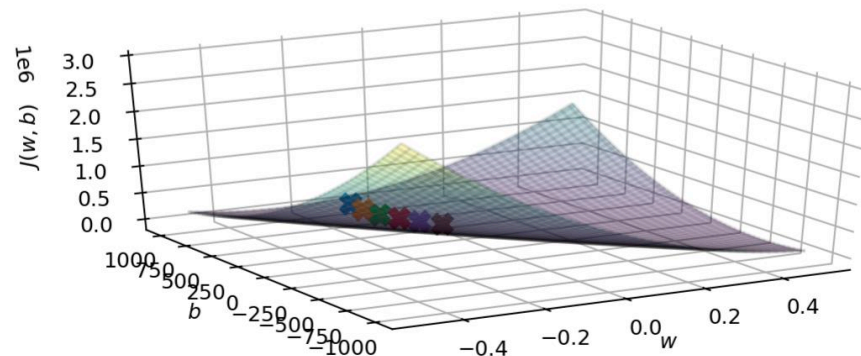
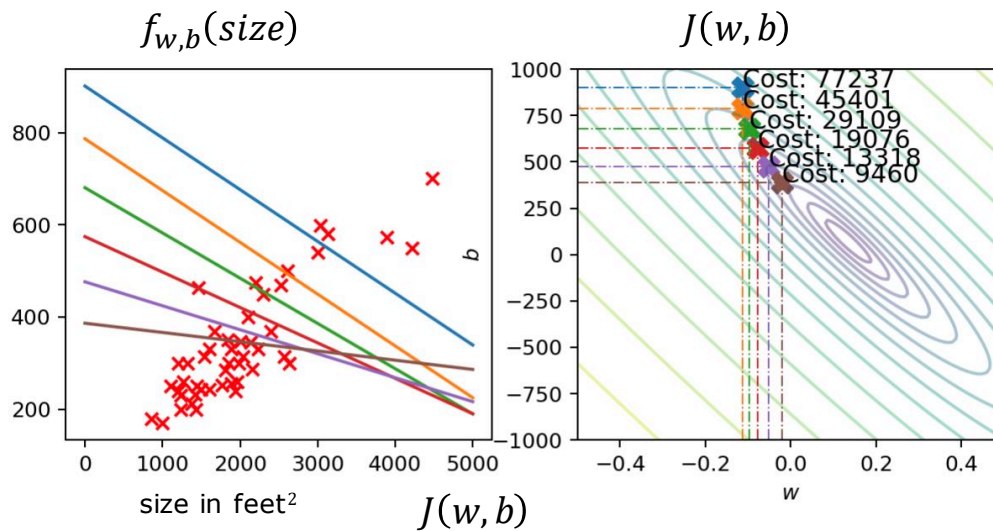
price in
\$1000's



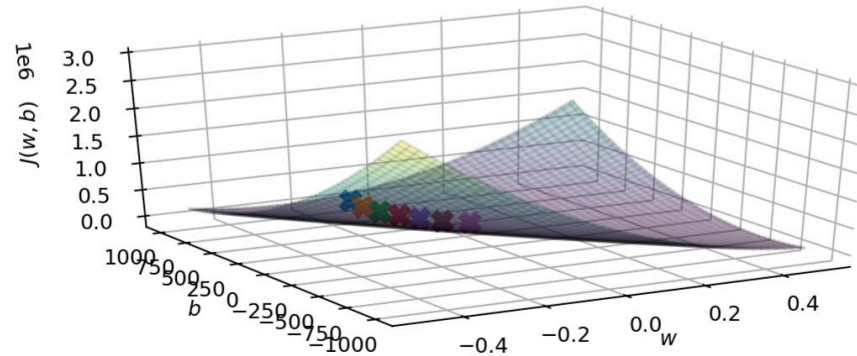
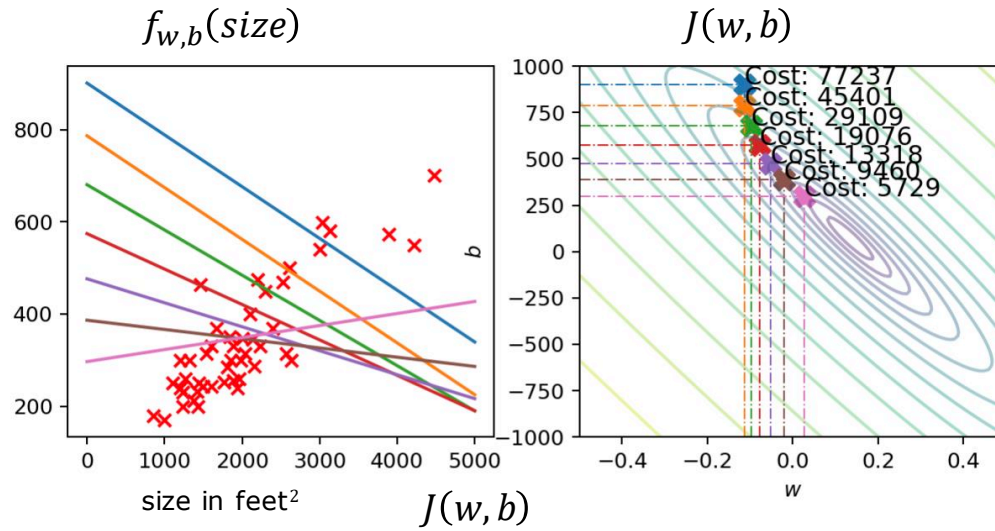
price in
\$1000's



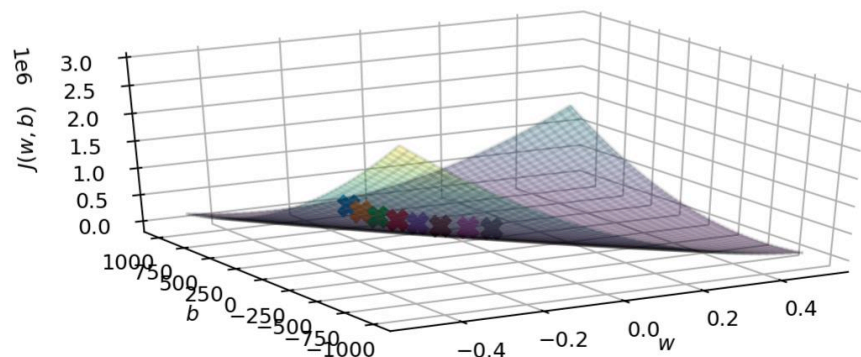
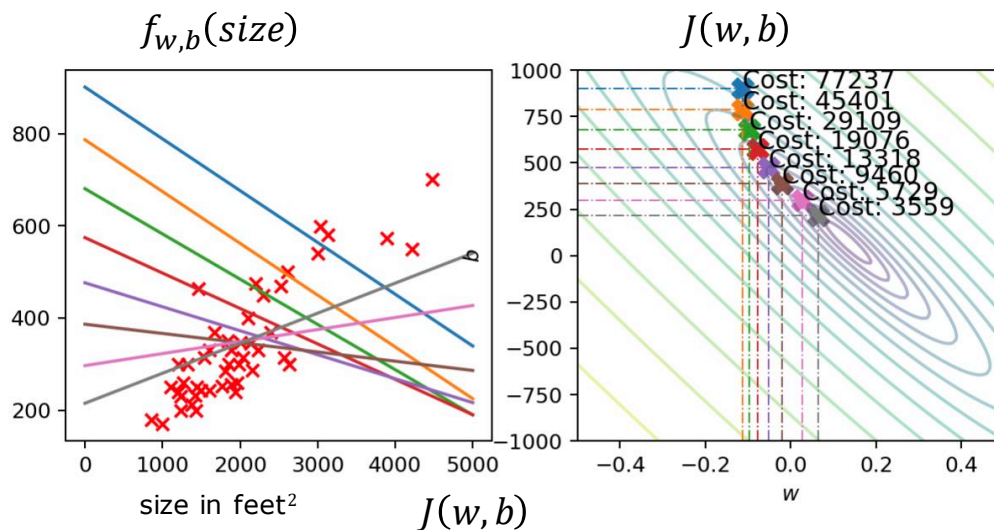
price in
\$1000's

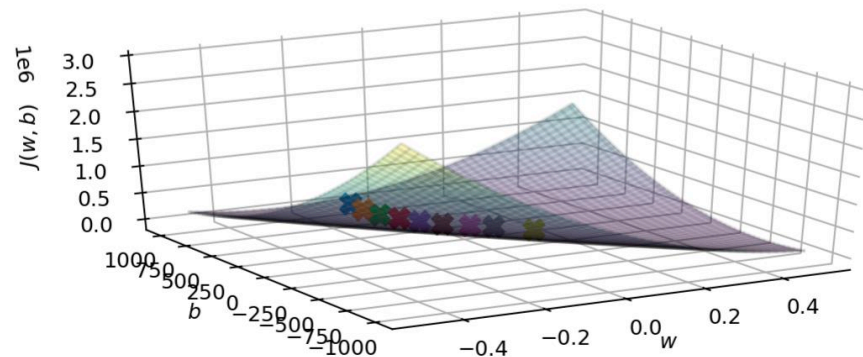
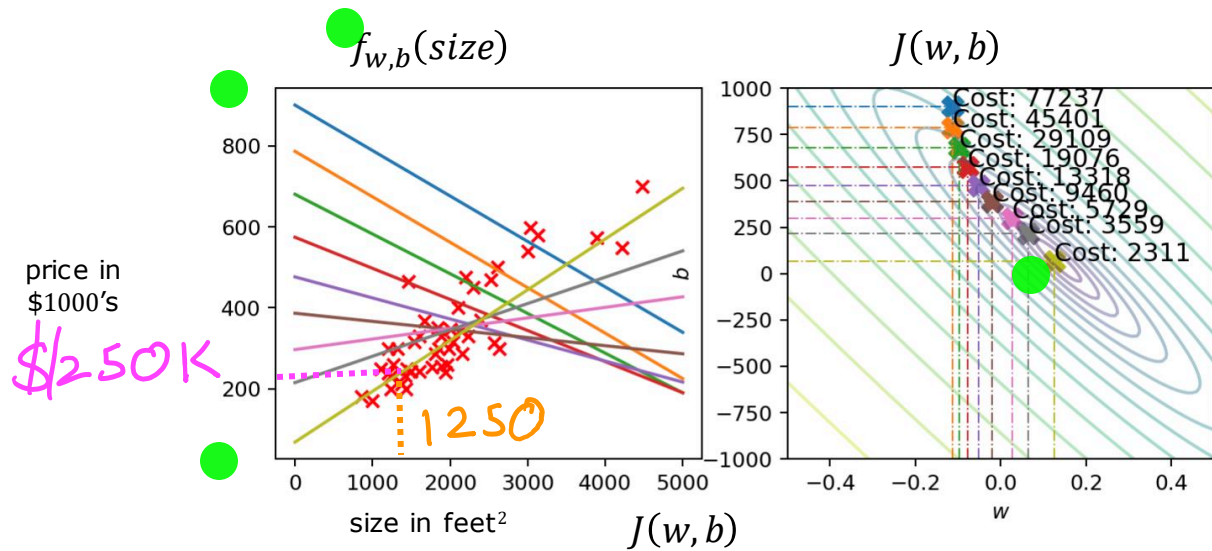


price in
\$1000's



price in
\$1000's





"Batch" gradient descent

"Batch": Each step of gradient descent uses all the training examples.

other gradient descent: subsets

	x size in feet ²	y price in \$1000's
(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
...	...	...
(47)	3210	870

$m = 47$

$$\sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

